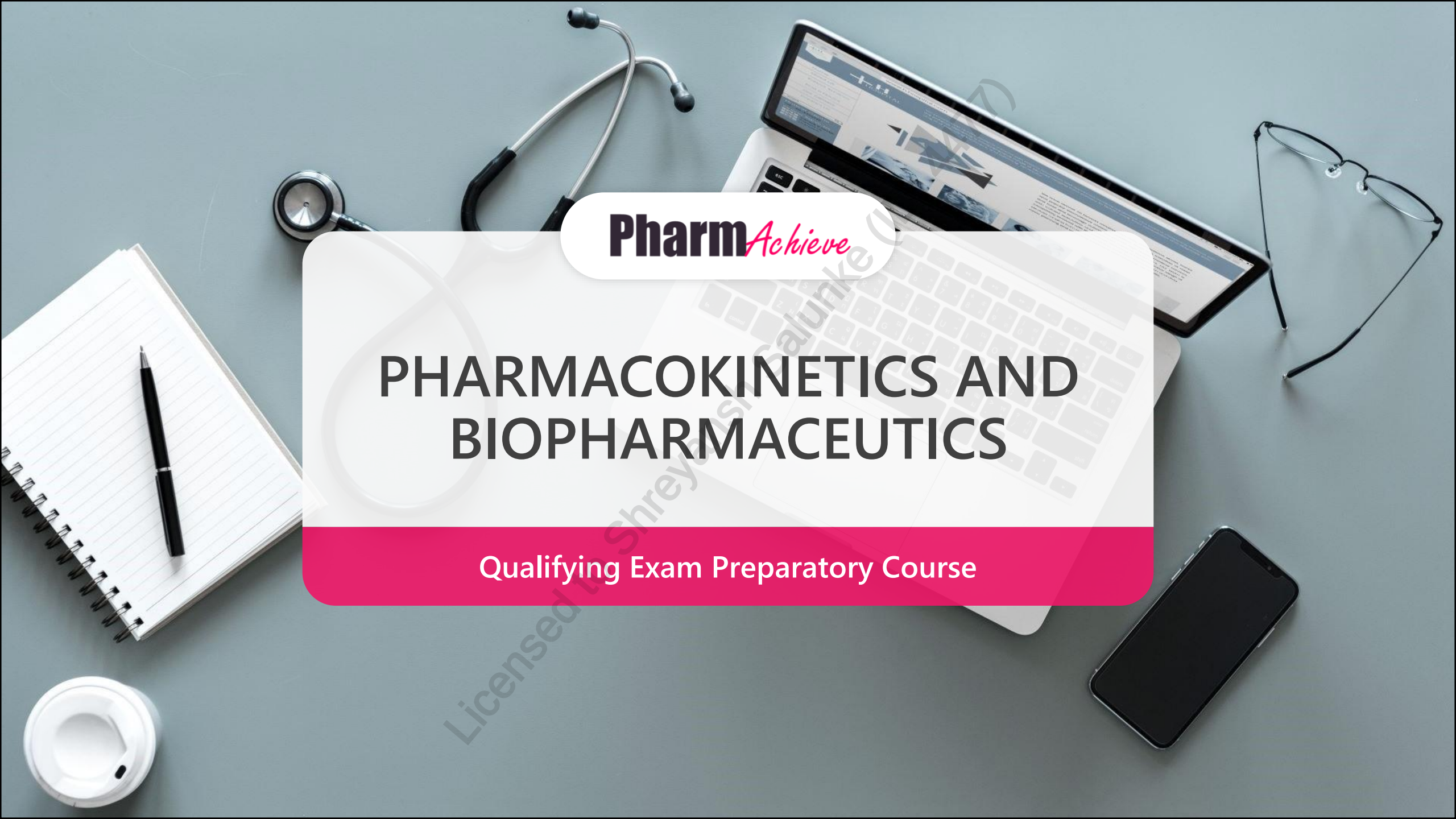




Pharm*Achieve*

PHARMACOKINETICS AND BIOPHARMACEUTICS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ADME: Absorption, Distribution, Metabolism, Excretion

ATP: Adenosine Triphosphate

AUC: Area Under The Curve

CL: Clearance

D: Dose

F: Bioavailability

IBW: Ideal Body Weight

IV: Intravenous

Kg: Kilogram

L: Liter

MAOI: Monoamine Oxidase Inhibitor

MEC: Minimum Effective Concentration

MIC: Minimum Inhibitory Concentration

OAT: Organic Anion Transporter

PD: Pharmacodynamics

P-gp: P-Glycoprotein

PK: Pharmacokinetics

PO: *Per os* (oral)

SSRI: Selective Serotonin Reuptake Inhibitor

TDM: Therapeutic Drug Monitoring

Vd: Volume of Distribution

τ : Dosing Interval

TABLE OF CONTENTS

Lecture	Topics
Pharmacokinetics Part 1	• Introduction to PK • Absorption • Distribution • Elimination • Metabolism • Compartment Models • IV bolus Drugs • Oral Drug Absorption
Pharmacokinetics Part 2	• Multiple Doses and Steady-State • Therapeutic Drug Monitoring • Non-linear PK • Interactions • Pharmacodynamics

EXPONENTS AND LOGARITHMS

Form = X^a

X is the base

A is the exponent

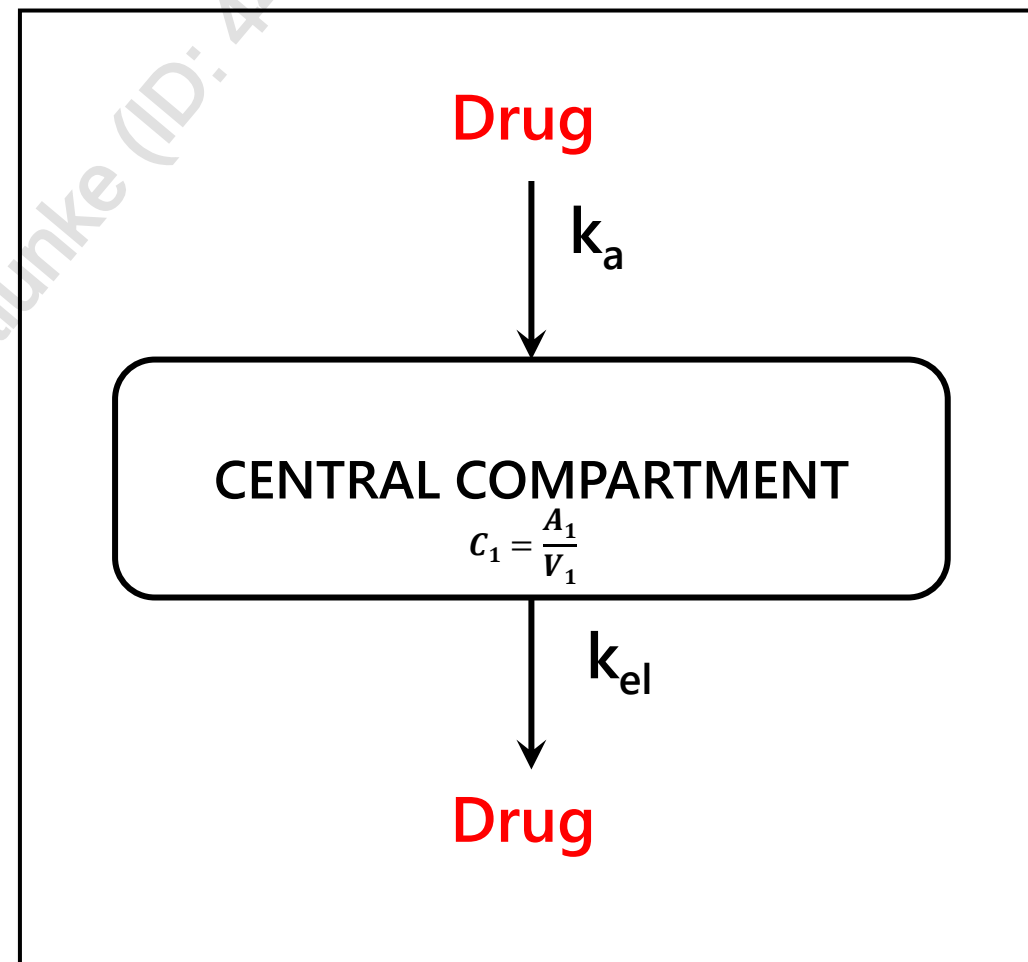
If $X^b = Y$, then $\log_X Y = b$

- $X^0 = 1, X^1 = X$
- $X^{-a} = \frac{1}{X^a}$
- $X^a * X^b = X^{a+b}$
- $\frac{X^a}{X^b} = X^{a-b}$
- $X^{a^b} = X^{a*b}$
- $(X \cdot Y)^a = X^a \cdot Y^a$

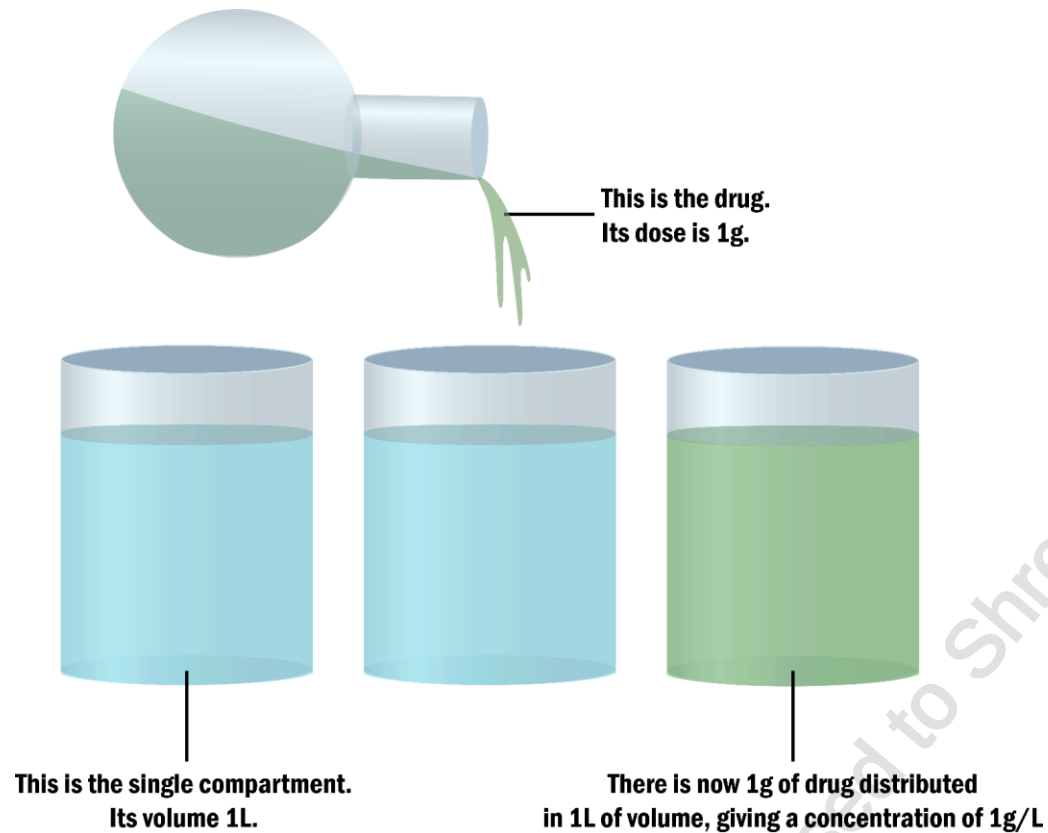
- $\log_a a^x = x$
- $\log(XY) = \log X + \log Y$
- $\log \frac{X}{Y} = \log X - \log Y$
- $\log(X^y) = y \log X$
- $\log_a X = \frac{\log_b X}{\log_b a}$
- $\log(X + Y) \neq \log X + \log Y$

ONE-COMPARTMENT MODEL

- One-compartment model views the entire body as a **single central compartment**
- Drug undergoes rapid uptake into all tissues it can enter - "Well-Stirred"
- Drug distribution is **instantaneous** in this model
- Compartment is characterized by values
 - V_1 is the volume and is equal to the drug's volume of distribution
 - A_1 is the amount of drug in the compartment and is equal to the amount of drug in the body
 - C_1 is the concentration and is equal to the drug plasma concentration
 - k_a = rate of absorption
 - k_{el} = rate of elimination



ONE COMPARTMENT MODEL



ONE-COMPARTMENT PLASMA CONCENTRATION-TIME CURVE

One Compartment Model

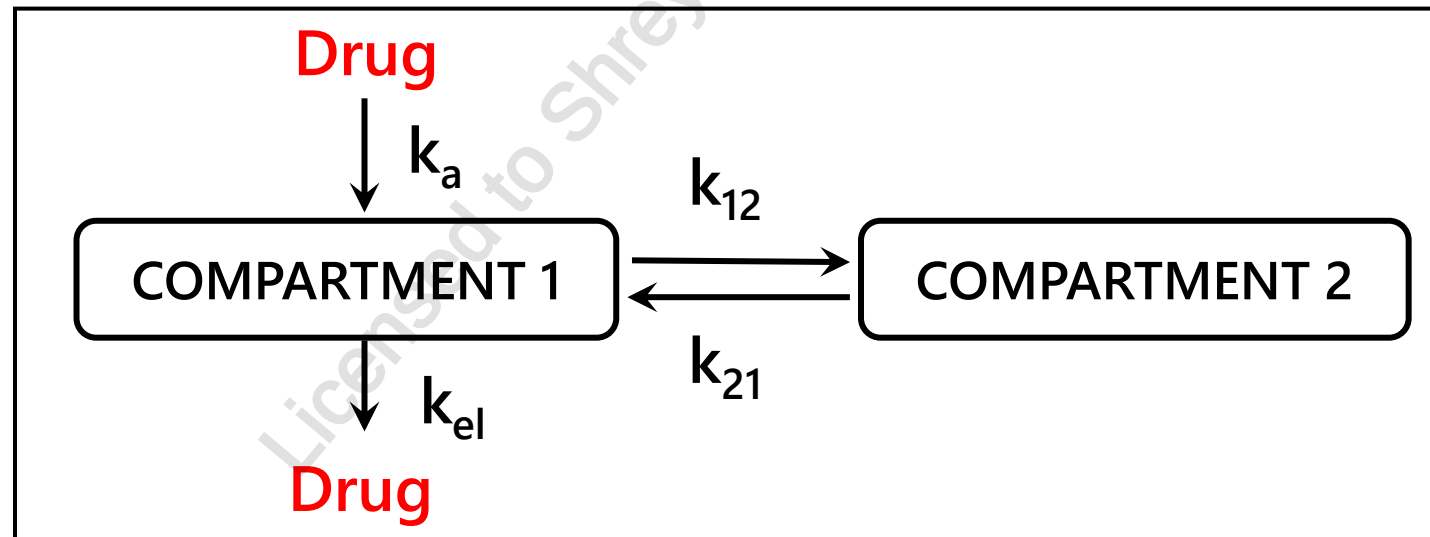
$\text{Log}(C_p)$

t

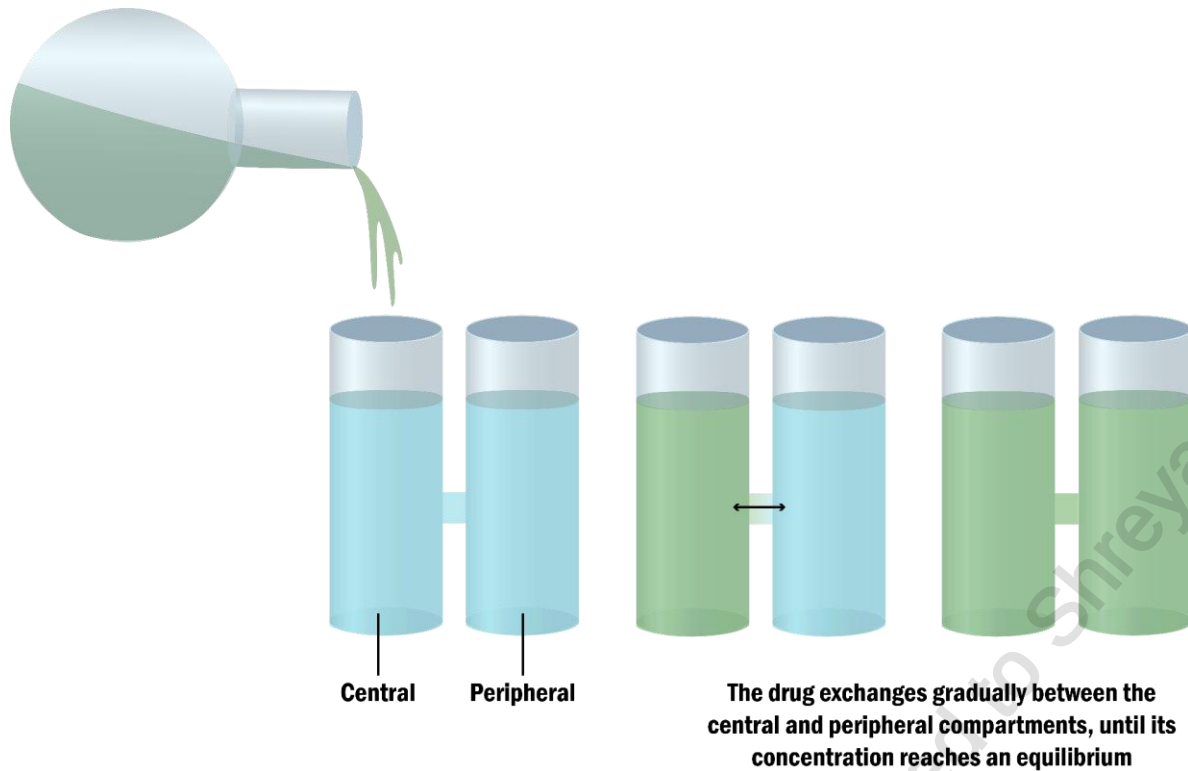
$$C_t = C_0 \cdot e^{-k_{el} \cdot t}$$
$$\ln(C_t) = \ln(C_0) - k_{el} \cdot t$$

TWO-COMPARTMENT MODEL

- Two-compartment model consists of the **central** compartment and a **peripheral** compartment
- Peripheral compartment consists of tissue where uptake is slower than central compartment
- Volume, drug amount, and concentration symbols for peripheral compartment denoted with a 2 (V_2 , A_2 , C_2)
- Distribution to peripheral compartment driven by A_1 , with rate constant k_{12}
- Redistribution to the central compartment is driven by A_2 , with rate constant k_{21}
- Rate constant for elimination is k_{10}

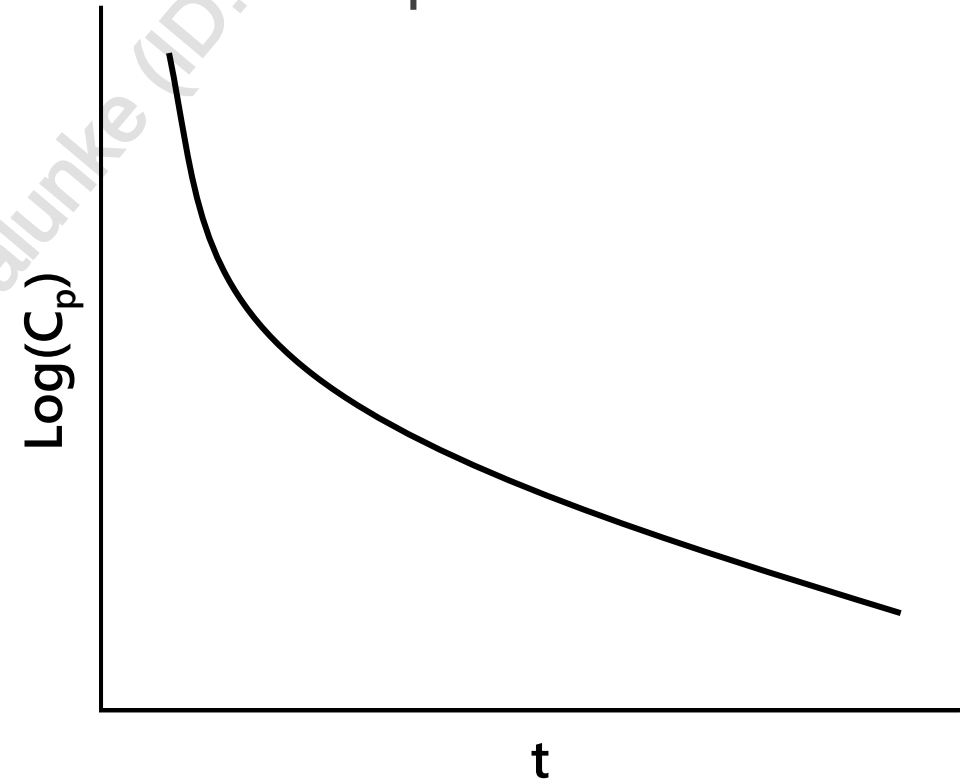


TWO COMPARTMENT MODEL



TWO-COMPARTMENT PLASMA CONCENTRATION-TIME CURVE

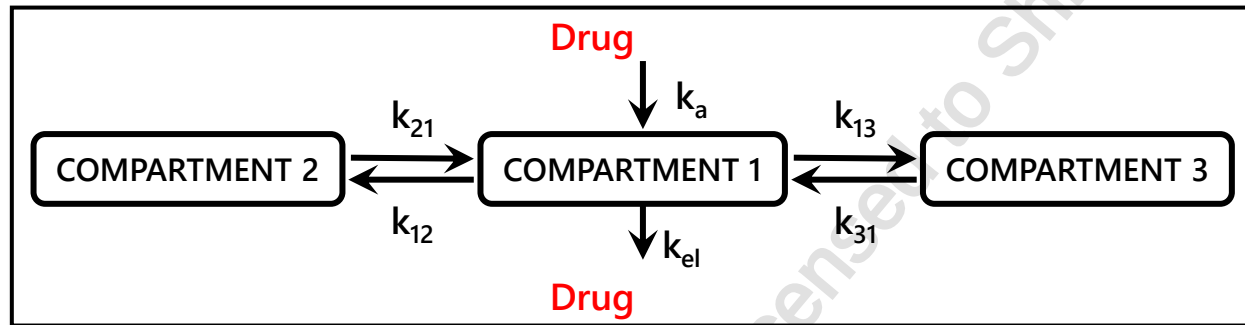
Two Compartment Model



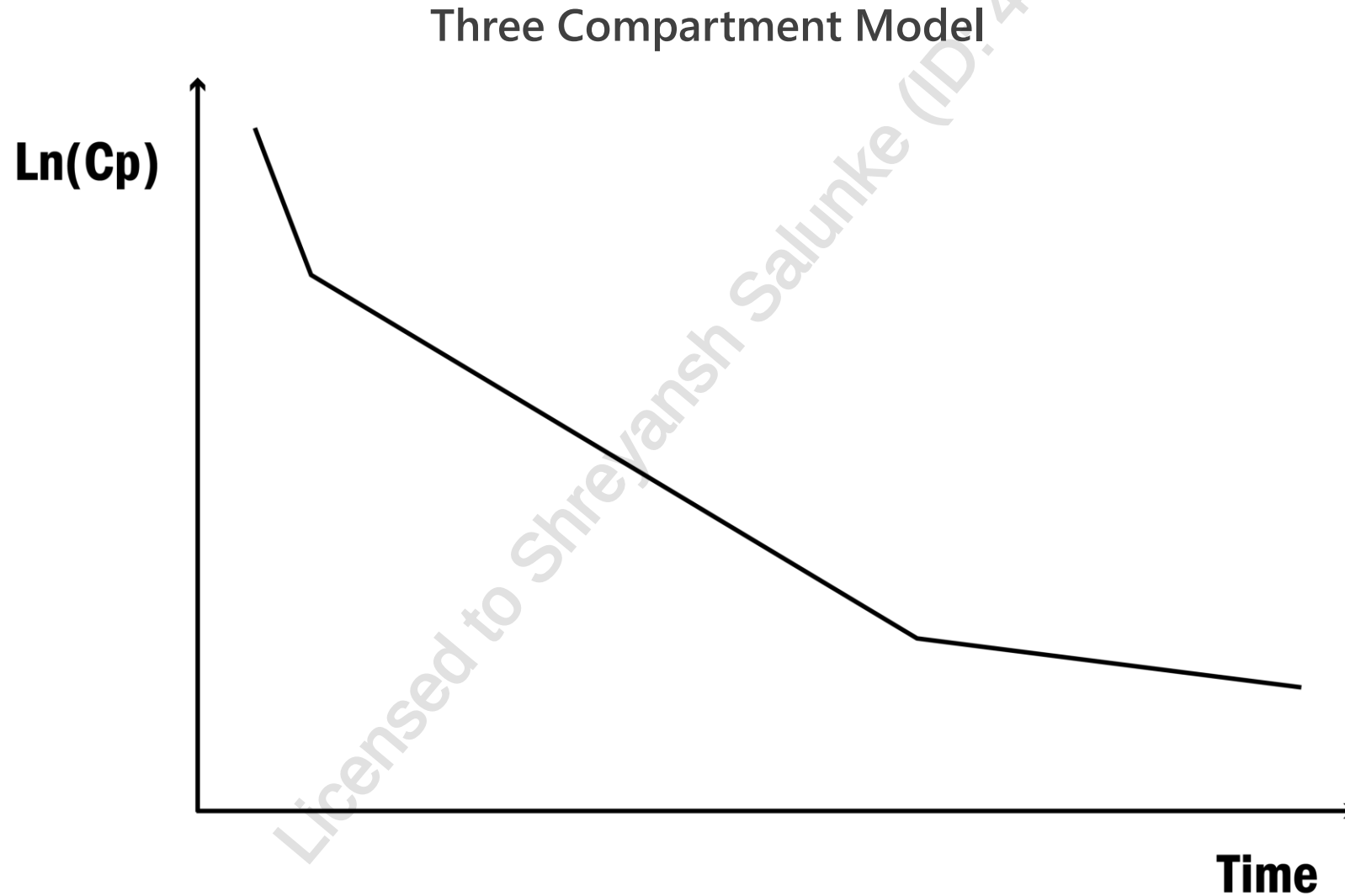
$$C_t = C_1 \cdot e^{-\alpha \cdot t} + C_2 \cdot e^{-\beta \cdot t}$$

THREE-COMPARTMENT MODEL

- Three-compartment model consists of the **central** compartment, **peripheral** compartment, and **deep tissue** compartment
- Deep tissue compartment consists of **poorly perfused tissue** that the drug distributes into very slowly
- Deep tissue volume, drug amount, and concentration denoted with a 3 (V_3 , A_3 , C_3)
- Distribution constant is k_{13} ; redistribution constant is k_{31}
- Rate of redistribution from deep tissue compartment **controls final stages of elimination**



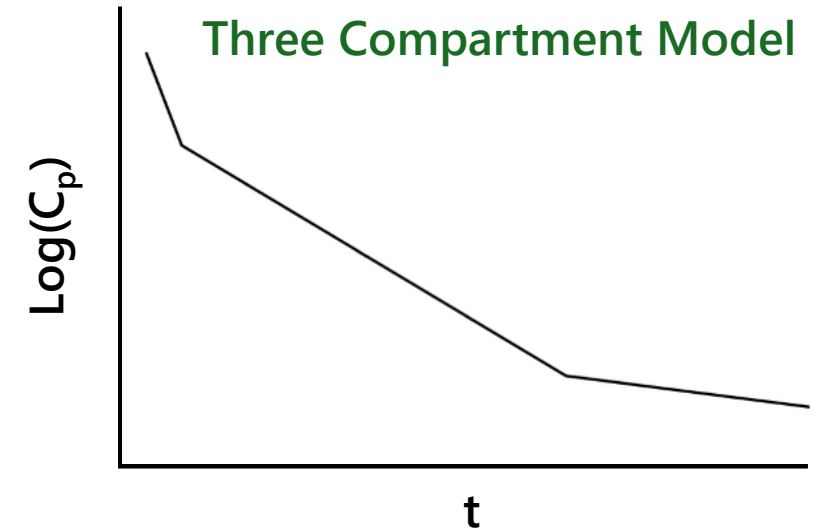
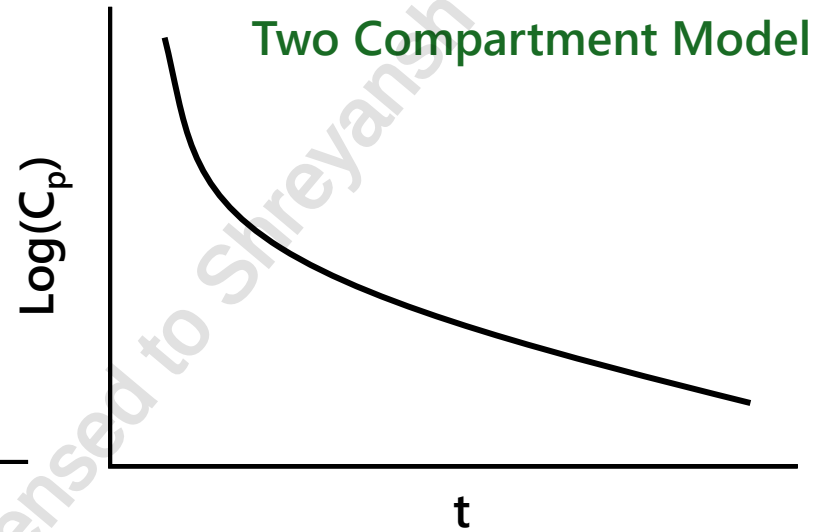
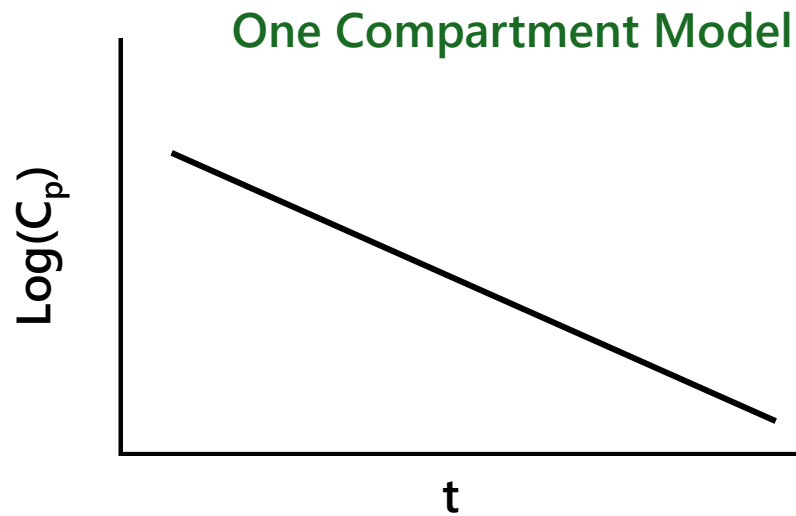
THREE-COMPARTMENT PLASMA CONCENTRATION TIME CURVE



DRUG CONCENTRATIONS VERSUS TIME CURVES

DETERMINING WHICH MODEL THE DRUG FOLLOWS

The number of slopes [phases] visible on a semi-log plot can be used to estimate the number of compartments a model should have



INTRAVENOUS INJECTION

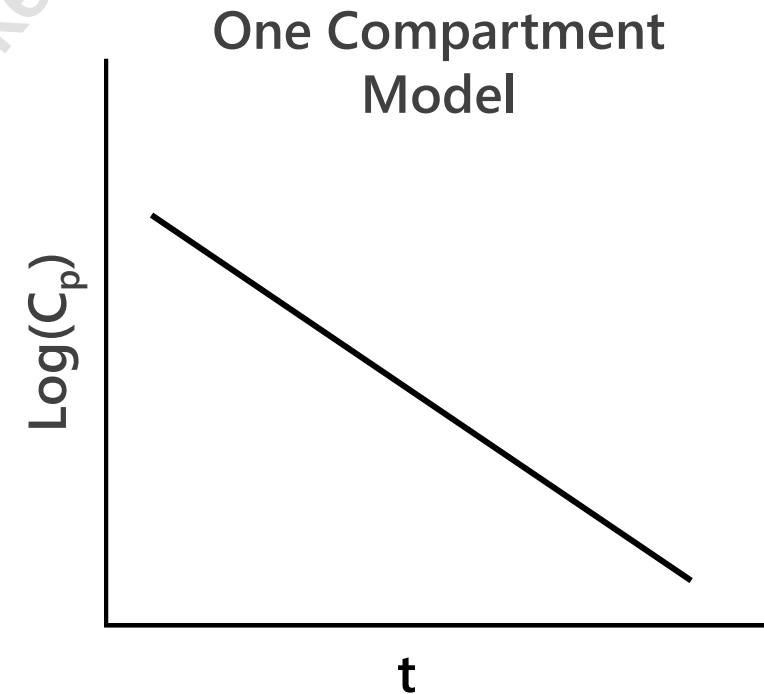
- A route of administration that **directly** puts drug into systemic circulation, providing **immediate** therapeutic plasma concentration (**recall $F = 100\%$!**)

Intravenous route of administration	
Bolus	Infusion
• Single dose administration of medication in a short period of time	• Dose that is administered over an extended period • Can be intermittent (e.g. at regular intervals) or continuous (e.g. at constant rate)

SOME IV DRUGS CAN BE MODELED AS 1-COMPARTMENT

- Elimination is the only process influencing plasma concentration
- Distribution equilibrium between plasma and tissue assumed to be instantaneous
- Log C_p vs. time forms a straight line
- Plasma concentration represents total body concentration
- Volume of compartment is equal to volume of distribution ($V_1 = V_d$)

Most drugs distribute into multiple compartments, but **when distribution is rapid relative to elimination** → they can be modeled as one compartment



IV BOLUS PLASMA CONCENTRATION

- The rate of drug over time at the start is equal to the difference between the rate of inputs and the rate of outputs

The plasma concentration is found by integrating the rate of drug over time from zero to infinity:

$$C_p = \frac{F \cdot D}{V_d} \cdot e^{-kt} = C_{p_0} \cdot e^{-kt}$$

- C_p is the plasma concentration
- V_d is the volume of distribution
- F is the bioavailability factor (=1 since IV)
- D is the dose administered
- C_{p_0} is the plasma concentration at time zero
- k is the elimination rate constant

SAMPLE PROBLEM

7. A 50-mg dose of a drug is administered via an intravenous injection. The drug has a volume of distribution of 100 L, and an elimination rate constant of 0.2 h^{-1} ?
- a) What is the initial plasma concentration?
 - b) What is the plasma concentration 4 hours after administration?

PRACTICE QUESTION 7

7. A 50-mg dose of a drug is administered via an intravenous injection. The drug has a volume of distribution of 100 L, and an elimination rate constant of 0.2 h⁻¹?
- a) What is the initial plasma concentration?

$$Cp_0 = \frac{F \cdot D}{Vd}$$

$$Cp_0 = \frac{1 \cdot 50 \text{ mg}}{100 \text{ L}}$$

$$Cp_0 = 0.5 \frac{\text{mg}}{\text{L}}$$

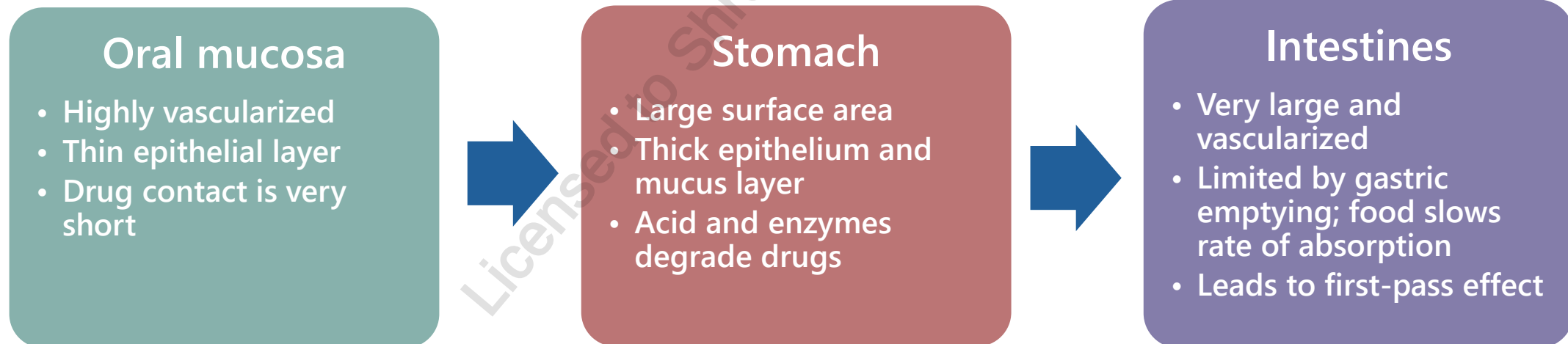
PRACTICE QUESTION 7- ANSWER

7. A 50-mg dose of a drug is administered via an intravenous injection. The drug has a volume of distribution of 100 L, and an elimination rate constant of 0.2 h⁻¹?
- b) What is the plasma concentration 4 hours after administration?

$$C_p = \frac{F \cdot D}{V_d} \cdot e^{-kt}$$
$$C_p = \frac{1 \cdot 50 \text{ mg}}{100 \text{ L}} \cdot e^{-0.2 \text{ h}^{-1} \cdot 4 \text{ h}}$$
$$C_p = 0.22 \frac{\text{mg}}{\text{L}}$$

ORAL ADMINISTRATION

- Most common form of administration
- Sites of absorption: oral mucosa, stomach and intestines
- Obstacles: stomach acid, proteases, poor diffusion, transit time
- Physiological considerations:
 - Absorption highly enhanced in thin, vascularized tissue
 - Acids can ionize many drugs, but can unionize drugs with a low pK_a
 - Food will often slow absorption, this may be beneficial for a sustained release
 - All drugs absorbed in the intestines are subject to the first-pass effect



PLASMA CONCENTRATION AFTER ORAL DOSE

The plasma concentration at any time after a dose can be found using the following biexponential equation:

$$C_p = \frac{F \cdot D \cdot k_a}{V_d \cdot (k_a - k)} \cdot (e^{-kt} - e^{-k_a t})$$

- C_p is the plasma concentration
- V_d is the volume of distribution
- k is the elimination rate constant
- F is the bioavailability
- D is the dose
- k_a is the absorption rate constant
- t is time

MODIFIED RELEASE PRODUCTS

- Immediate release of a drug is not always viable or desired; drug release rate and/or timing can be controlled
- Allows for greater compliance (e.g. decreased frequency of administration)
- Can help targeting and protect drugs; increases utility and range of oral drugs

Extended-release drugs

- Drugs are released over an extended period of time
- Useful with drugs that are cleared quickly and have targeted bioavailability
- Repeat action: releasing a dose at constant intervals

Delayed release drugs

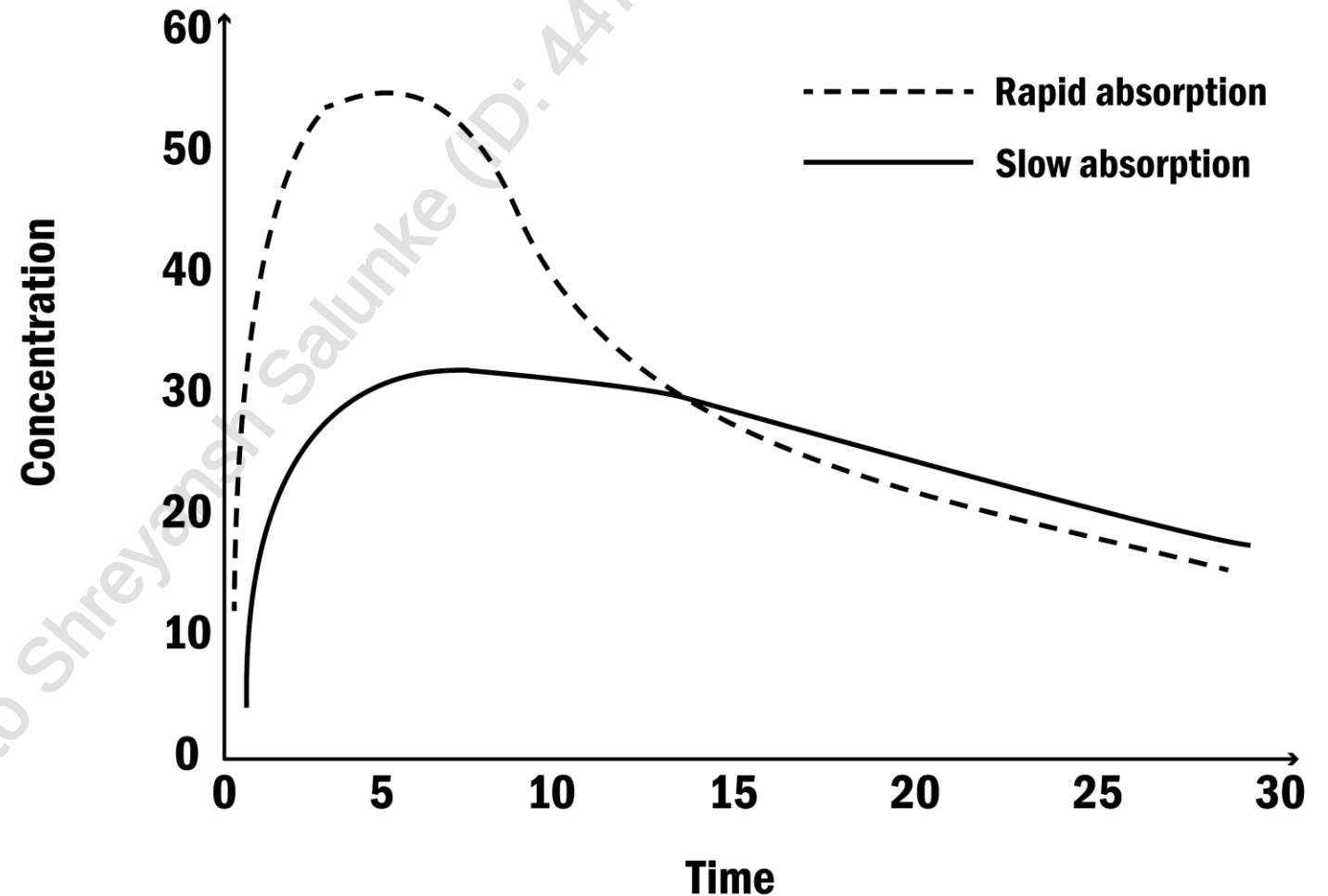
- Coatings prevent immediate release of drugs; often used to prevent drug-stomach interactions (e.g. acetylsalicylic acid)

Target release drugs

- Drugs packed in carriers such as liposomes, micelles, etc.
- Carriers are absorbed and transported to targets
- Allows for precise targeting of harmful drugs

FLIP-FLOP KINETICS

- The terminal fall in plasma concentration is controlled by k_a
- This is the flip-flop model
- Most prolonged-release preparations follow the flip-flop model
- Flip-flop kinetics occur when **the absorption rate is much slower than the elimination rate**



ACCUMULATION AND FLUCTUATION

- Maintenance of drug therapy involve repeated administration (i.e. **repeated doses of a drug at regular intervals**)
- **Accumulation**: gradual buildup of drug concentrations with each successive dose
 - Accumulation **occurs when a new dose is administered before the previous dose has been eliminated**
 - Fluctuation: **rise and fall** in plasma concentration during repeated dosing
 - Peak concentration is seen **at time of or right after** dose administration
 - Trough concentration is seen at time **right before** the next dose

INTRAVENOUS INFUSION

- To understand the PK of multiple doses, consider a **zero-order continuous drug input** into the body:
 - **Continuous drug input** into the bloodstream
 - **No fluctuation** in plasma concentration
 - Maintains **constant plasma concentration**
 - Very **accurate means** of drug administration

INTRAVENOUS INFUSION BASIC EQUATION

The rate of change of the amount of drug in the body is equal to the difference between the rate of inputs and the rate of outputs

$$Cp = \frac{R_0}{Cl} \cdot (1 - e^{-kt})$$

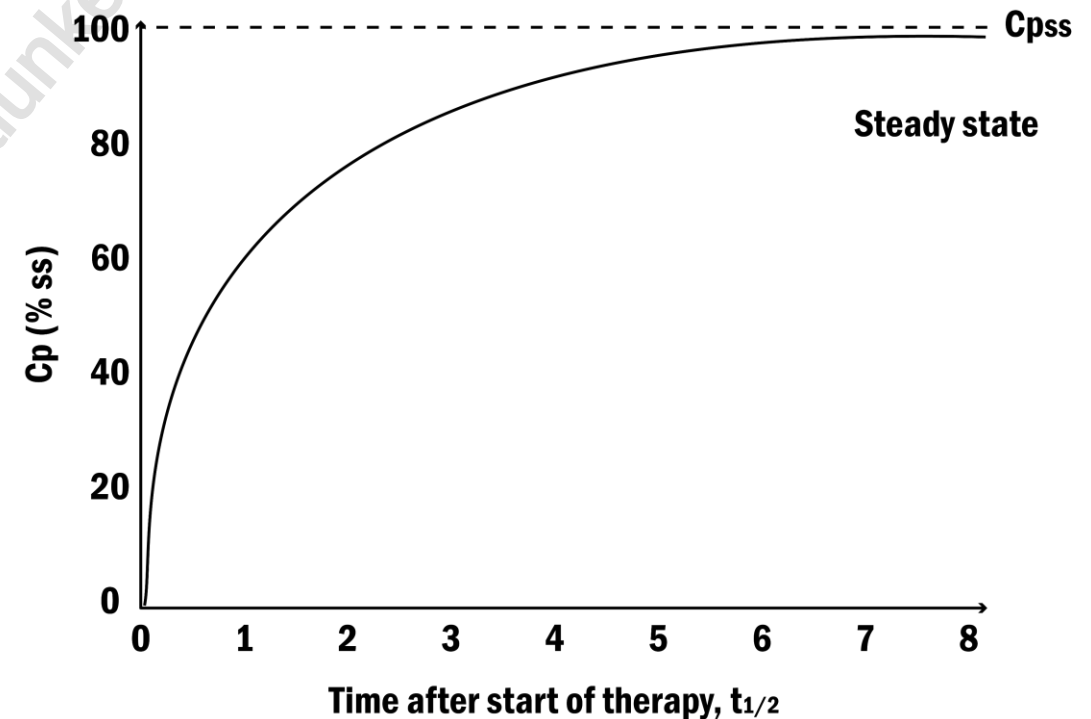
- **Cp** is the plasma concentration
- **Cl** is the drug clearance
- **R₀** is the infusion rate
- **k** is the elimination rate constant
- **t** is the time

IMPORTANT PROPERTIES OF THE BASIC EQUATION

- We can derive several important properties from the basic equation
 1. The **growth factor is $1 - e^{-kt}$** ; starts at zero when $t = 0$ and goes to 1 when $t = \infty$
 2. Value of **k determines the growth speed**
 3. **Plasma concentration at any time can be found** during zero-order drug input
 4. Approximately **5 elimination half-lives until steady-state** is reached
 5. During constant IV infusion, drug **accumulation is a self-limiting process** that stops at steady state

STEADY-STATE IV INFUSION

- **Steady-state:** plasma concentration constant and unaffected by time (**rate in = rate out**)
- Steady-state plasma concentration aimed for the middle of the therapeutic range
- The **growth factor tends towards 1** and the **plasma concentration tends towards the steady-state concentration**
- Several important properties of steady state
 1. Proportional to infusion rate
 2. Inversely proportional to drug's clearance
 3. Independent of volume of distribution
 4. Growth factor: the fraction of steady state achieved at any time



IV INFUSION STEADY-STATE FORMULA

Steady-state is achieved when the growth factor is equal to 1

$$Cp_{ss} = \frac{R_0}{Cl}$$

- Cp_{ss} is the steady-state plasma concentration
- Cl is the drug clearance
- R_0 is the constant rate of administration

LOADING DOSE

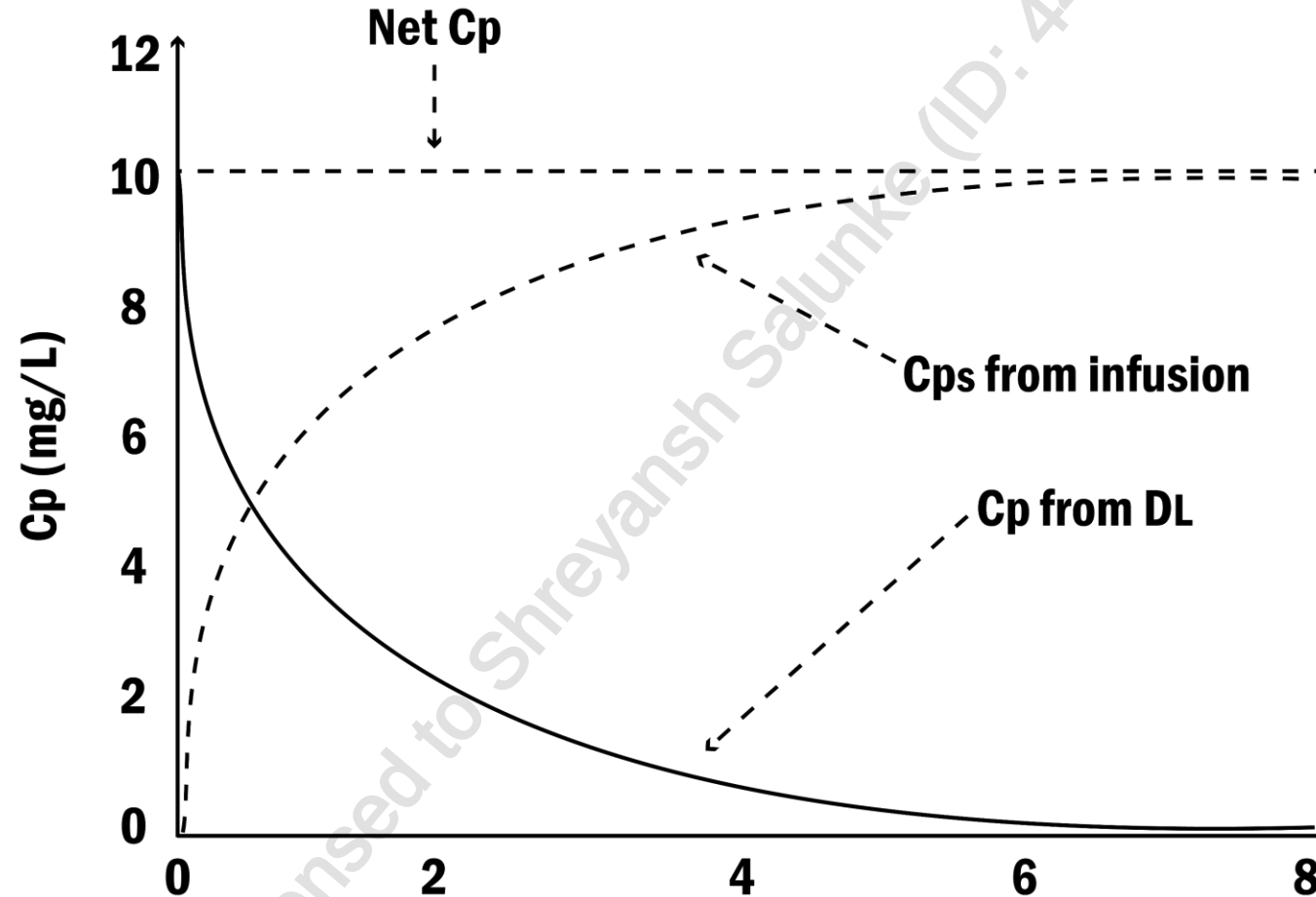
A loading dose can be used if a drug has a long half-life or therapeutic concentrations are needed immediately

The loading dose can be calculated by using the steady-state as the target:

$$D_L = \frac{Cp_{ss} \cdot Vd}{S \cdot F}$$

- Cp_{ss} is the steady state plasma concentration
- S is the salt factor
- F is the bioavailability factor
- Vd is the volume of distribution
- D_L is the loading dose

LOADING DOSE AND INFUSION RATES



Time after loading dose and start of infusion,

Note: $t_{1/2} = 1$ h

PRACTICE QUESTION 8

8. A drug ($Cl = 18 \text{ L/h}$, $V_d = 140 \text{ L}$) is administered by IV infusion at a rate of 10 mg/h .
- a) What is the plasma concentration 6 hours after infusion start?
 - b) What is the steady state plasma concentration?
 - c) Calculate the plasma concentration 4 hours after termination if the infusion was stopped 6 hours from the start.

PRACTICE QUESTION 8- ANSWER

8. A drug ($Cl = 18 \text{ L/h}$, $V_d = 140 \text{ L}$) is administered by IV infusion at a rate of 10 mg/h .
- a) What is the plasma concentration 6 hours after infusion start?

$$k = \frac{Cl}{V_d}$$

$$k = \frac{18 \frac{\text{L}}{\text{h}}}{140 \text{ L}}$$

$$k = 0.13 \text{ h}^{-1}$$

$$C_p = \frac{R_0}{Cl} \cdot (1 - e^{-kt})$$

$$C_p = \frac{10 \frac{\text{mg}}{\text{h}}}{18 \frac{\text{L}}{\text{h}}} \cdot (1 - e^{-0.13 \text{ h}^{-1} \cdot 6 \text{ h}})$$

$$C_p = 0.30 \frac{\text{mg}}{\text{L}}$$

PRACTICE QUESTION 8- ANSWER

8. A drug ($Cl = 18L/h$, $Vd = 140 L$) is administered by IV infusion at a rate of 10 mg/h .
b) What is the steady state plasma concentration?

$$Cp_{ss} = \frac{R_0}{Cl}$$

$$Cp_{ss} = \frac{10 \frac{mg}{h}}{18 \frac{L}{h}}$$

$$Cp_{ss} = 0.56 \frac{mg}{L}$$

PRACTICE QUESTION 8- ANSWER

8. A drug ($Cl = 18 \text{ L/h}$, $V_d = 140 \text{ L}$) is administered by IV infusion at a rate of 10 mg/h .
- c) Calculate the plasma concentration 4 hours after termination if the infusion was stopped 6 hours from the start.

$$C_p = C_{pT} \cdot e^{-kt'}$$

$$C_p = 0.30 \frac{\text{mg}}{\text{L}} \cdot e^{-kt'}$$

$$C_p = 0.30 \frac{\text{mg}}{\text{L}} \cdot e^{-0.13 \cdot 4}$$

$$C_p = 0.18 \frac{\text{mg}}{\text{L}}$$

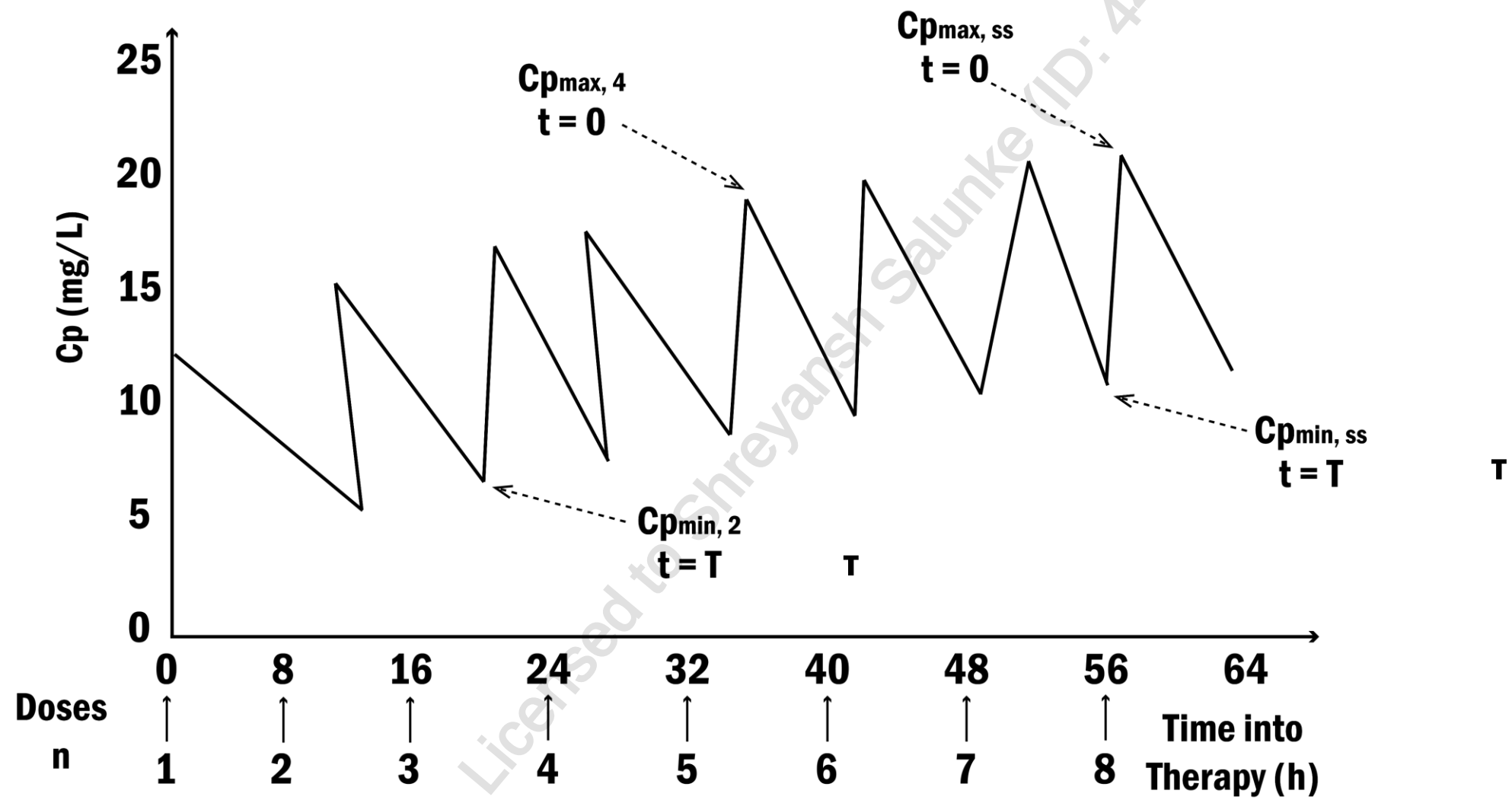
MULTIPLE INTRAVENOUS BOLUS INJECTIONS

- Goal of drug therapy is to **maintain a therapeutic concentration**
- Several vital qualities in a plasma concentration-time profile for multiple intravenous doses
 1. Fluctuation
 2. Accumulation
 3. Steady-state
 4. Average C_p
- Several assumptions that need to be made for multiple-doses
 - Except for the loading dose, the **dose and dosing interval are constant**. Dose is known as maintenance dose
 - **PK parameters remain constant** during therapy; linear kinetics
 - When a dose is administered, it is assumed the **entire dose enters systemic circulation at time zero**

IMPORTANT ASPECTS OF THE MULTIPLE DOSING EQUATION

- Several important terms we need to define
 - n is the number of the last dose given
 - τ is the dosing interval
 - t is the time since the last dose
 - Time into therapy is the product of n and τ
- Considerations
 - When $t = 0$, Cp_{max} (recall that entire dose enters circulation)
 - When $t = \tau$, Cp_{min} (end of the dosing interval, before next dose)
 - Cp_{max} and Cp_{min} depend on the # of the last dose, before steady state
 - After steady state, Cp_{max} and Cp_{min} remain constant with each dose

PLASMA CONCENTRATION-TIME PROFILE



STEADY-STATE BASIC EQUATION

The steady-state basic equation is:

$$Cp_{ss} = \frac{D}{Vd \cdot (1 - e^{-k\tau})} \cdot e^{-kt}$$

- Cp_{ss} is the plasma concentration
- k is the elimination rate constant
- τ is the dosing interval
- Vd is the volume of distribution
- D is the dose
- t is the time

STEADY-STATE BASIC EQUATION

The steady-state basic equation for the maximum plasma concentration:

$$Cp_{max,ss} = \frac{D}{Vd \cdot (1 - e^{-k\tau})}$$

- $Cp_{max,ss}$ is the maximum plasma concentration at steady state
- k is the elimination rate constant
- τ is the dosing interval
- Vd is the volume of distribution
- D is the dose

STEADY-STATE BASIC EQUATION

The steady-state basic equation for the minimum plasma concentration:

$$Cp_{min,ss} = Cp_{max,ss} \cdot e^{-k\tau}$$

- $Cp_{max,ss}$ is the maximum plasma concentration at steady state
- k is the elimination rate constant
- τ is the dosing interval
- $Cp_{min,ss}$ is the minimum steady-state plasma concentration

AVERAGE PLASMA CONCENTRATION

The average plasma concentration is calculated by dividing the area under the steady state dosing interval and dividing it by the dosing interval:

$$Cp_{av,ss} = \frac{AUC_0^{\tau}}{\tau} = \frac{F \cdot D}{Cl \cdot \tau} = \frac{F \cdot R_a}{Cl}$$

- $Cp_{av,ss}$ is the average plasma concentration at steady state
- τ is the dosing interval
- R_a is the average rate of drug administration over the dosing interval
- F is the bioavailability factor
- Cl is the clearance
- D is the dose

FLUCTUATION

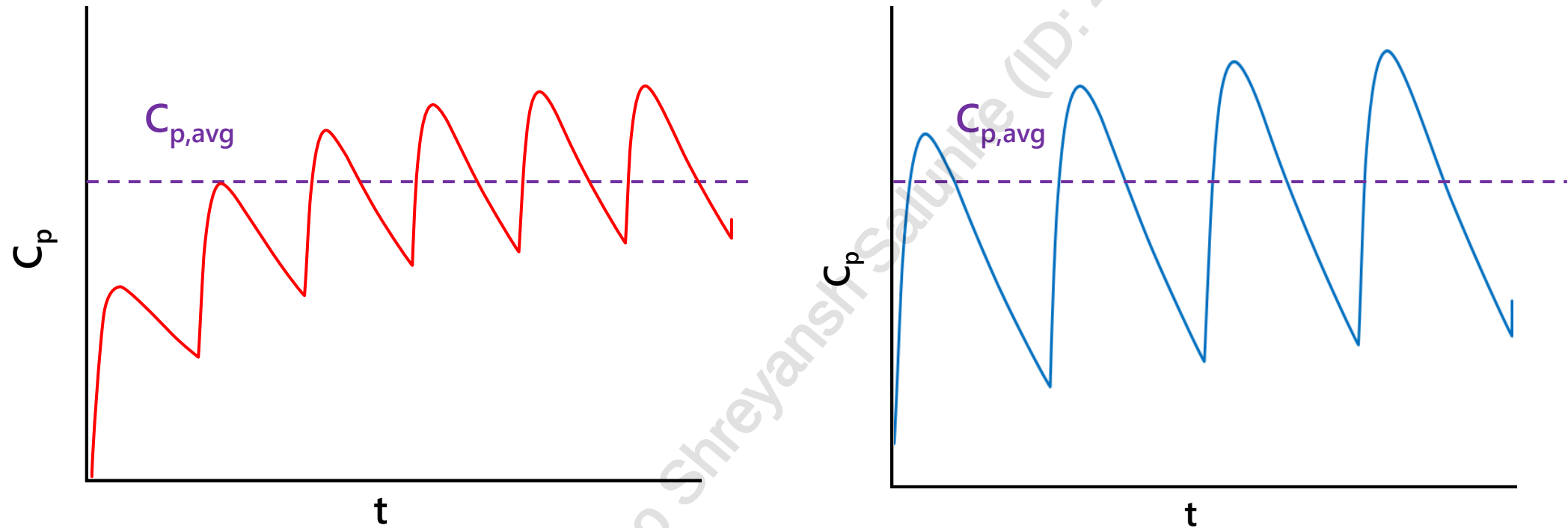
- **Fluctuation** is the **difference between a peak and a trough during a single dosing interval** and reflects the amount of drug eliminated during this interval
- **Elimination increases = fluctuation increases**
- Fluctuation is **dependent on the rate of elimination** and the **dosing interval**

Fluctuation is the ratio between the maximum and minimum plasma concentration at steady state:

$$\frac{Cp_{max,ss}}{Cp_{min,ss}} = \frac{1}{e^{-k\tau}}$$

- $Cp_{max,ss}$ is the **maximum plasma concentration at steady state**
- k is the **elimination rate constant**
- τ is the **dosing interval**
- $Cp_{min,ss}$ is the **minimum steady-state plasma concentration**

MULTIPLE DOSES



Lengthening the dosage interval while increasing the dosage by the same proportion, allows maintenance of the $C_{p,avg}$ but reduces min and increases max C_p , leading to a higher variation

ACCUMULATION

- **Accumulation** is the **increase** that occurs in **drug concentration** with each additional dose
- **Accumulation decreases until steady-state is achieved**
- **Accumulation ratio (r)** is the ratio of a drug going from a single dose to steady state with repeated administration

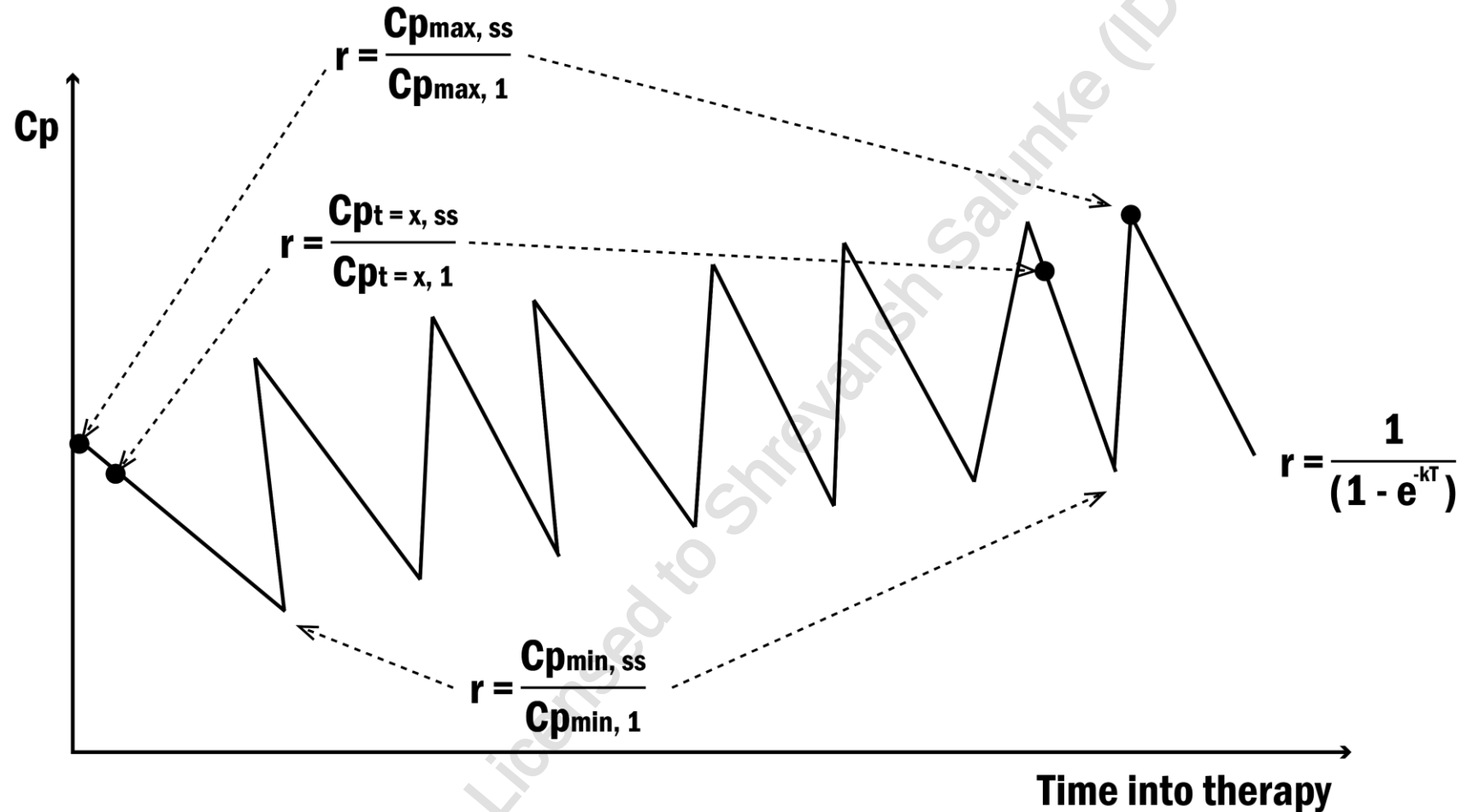
The accumulation ratio is found by comparing steady-state plasma concentrations with the equivalent concentration after one dose

$$r = \frac{Cp_{ss}}{Cp_1} = \frac{1}{1 - e^{-k\tau}}$$

- **r** is the **accumulation ratio**
- **Cp_{ss}** is the **plasma concentration at steady state**
- **k** is the **elimination rate constant**
- **τ** is the **dosing interval**
- **Cp₁** is the **plasma concentration after one dose**

ACCUMULATION

r = ratio of C_p at steady state to C_p at same time after the first dose



LOADING DOSE

A loading dose is used to achieve steady-state concentration immediately. To get to steady-state, we need r times more of the drug than the usual maintenance dose

$$D_L = \frac{D_M}{1 - e^{-k\tau}}$$

- D_L is the loading dose
- k is the elimination rate constant
- τ is the dosing interval
- D_M is the maintenance dose

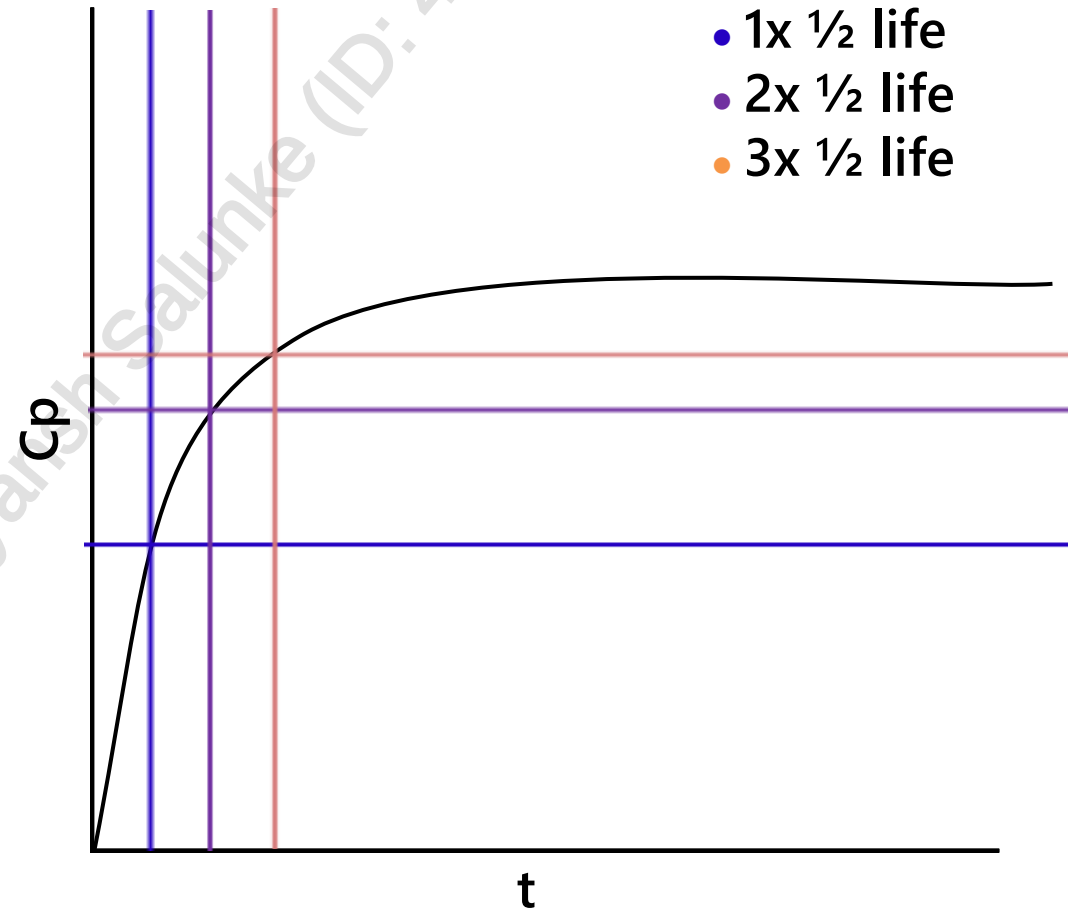
LOADING VS. MAINTENANCE DOSES

Loading Dose	Maintenance Doses
• When an immediate drug effect is needed • Parenteral route more common, but may be done orally as well • Rationale: give large dose of medication to “fill-up” the volume of distribution, and rapidly achieve therapeutic plasma concentration	• May be started after loading dose or alone if rapid effect not required • Steady-state levels achieved after ~5 half-lives • Give as either continuous infusion OR scheduled intermittent doses

TIME TO STEADY STATE

- Technically, C_p^{ss} is the C_p at $t = \infty$
- Time to $\frac{1}{2} C_p = t_{1/2} = \ln(2)/k_{el}$
- The $t_{1/2}$ is determined by rate of elimination only, k determines C_p^{ss}
- Note: The time to C_p^{ss} is exponential in nature. $\frac{1}{2}$ of the remaining distance is covered every $\frac{1}{2}$ life

# of half-lives	% of C_p^{ss} reached
1	50%
2	75%
3	87.5%
4	94%
5	97.25%



PRACTICE QUESTION 9

9. A synthetic analgesic has the following parameters: $V_d = 1.5 \text{ L/Kg}$, $Cl = 30 \text{ L/h}$; therapeutic range $0.5 - 1.5 \text{ mg/L}$. The drug is to administered as multiple IV doses.
- a) Determine an appropriate dosing regimen for a 75-kg male.
 - b) Calculate a suitable loading dose.
 - c) What is the average plasma concentration during steady-state?
 - d) After 10 days, a final dose is given, and the regimen is stopped.
 - i) How long will it approximately take for the drug to be completely eliminated?
 - ii) What is the plasma concentration 10 hours after the final dose is administered?

PRACTICE QUESTION 9- ANSWER

9. A synthetic analgesic has the following parameters: $V_d = 1.5 \text{ L/Kg}$, $Cl = 30 \text{ L/h}$; therapeutic range $0.5 - 1.5 \text{ mg/L}$. The drug is to administered as multiple IV doses.
- a) Determine an appropriate dosing regimen for a 75-kg male.

$$t_{1/2} = \frac{0.693 \cdot V_d}{Cl}$$

$$t_{1/2} = \frac{0.693 \cdot 75 \cdot 1.5}{30}$$

$$t_{1/2} = \mathbf{2.60 \text{ h}}$$

$$\tau = -\frac{1}{k} \cdot \ln \frac{Cp_{min,ss}}{Cp_{max,ss}}$$

$$\tau = -\frac{1}{0.267} \cdot \ln \frac{0.5 \frac{mg}{L}}{1.5 \frac{mg}{L}}$$

$$\tau = \mathbf{4.12 \text{ h}}$$

$$Cp_{max,ss} = \frac{D}{V_d \cdot (1 - e^{-k\tau})}$$

$$D = Cp_{max,ss} \cdot V_d \cdot (1 - e^{-k\tau})$$

$$D = 1.5 \cdot 75 \cdot 1.5 \cdot (1 - e^{-\frac{0.693}{2.60} \cdot 4.12})$$

$$D = \mathbf{113 \text{ mg}}$$

PRACTICE QUESTION 9- ANSWER

9. A synthetic analgesic has the following parameters: $V_d = 1.5 \text{ L/Kg}$, $Cl = 30 \text{ L/h}$; therapeutic range $0.5 - 1.5 \text{ mg/L}$
- b) Calculate a suitable loading dose.

$$D_L = \frac{D_M}{1 - e^{-k\tau}}$$
$$D_L = \frac{113}{1 - e^{-\frac{0.693}{2.60} \cdot 4.12}}$$
$$D_L = \mathbf{170 \text{ mg}}$$

PRACTICE QUESTION 9- ANSWER

9. A synthetic analgesic has the following parameters: $V_d = 1.5 \text{ L/Kg}$, $Cl = 30 \text{ L/h}$; therapeutic range $0.5 - 1.5 \text{ mg/L}$
- c) What is the average plasma concentration during steady-state?

$$Cp_{av,ss} = \frac{D}{Cl \cdot \tau}$$
$$Cp_{av,ss} = \frac{113 \text{ mg}}{30 \frac{\text{L}}{\text{h}} \cdot 4.12 \text{ h}}$$
$$Cp_{av,ss} = 0.91 \frac{\text{mg}}{\text{L}}$$

PRACTICE QUESTION 9- ANSWER

9. A synthetic analgesic has the following parameters: $V_d = 1.5 \text{ L/Kg}$, $Cl = 30 \text{ L/h}$; therapeutic range $0.5 - 1.5 \text{ mg/L}$
- d) After 10 days, a final dose is given, and the regimen is stopped.
- How long will it approximately take for the drug to be completely eliminated?
 - What is the plasma concentration 10 hours after the final dose is administered?

$$t_{1/2} \cdot 5 = 2.60 \text{ hours} \cdot 5 = \mathbf{13 \text{ hours}}$$

$$Cp = Cp_{max,ss} \cdot e^{-kt}$$

$$Cp = 1.5 \frac{\text{mg}}{\text{L}} \cdot e^{-\frac{0.693}{2.6 \text{ h}} \times 10 \text{ h}}$$

$$Cp = \mathbf{0.1 \frac{\text{mg}}{\text{L}}}$$

THERAPEUTIC DRUG MONITORING

CHARACTERISTICS OF DRUGS REQUIRING TDM

Predictable PK/PD relationship

Narrow therapeutic window

Adjustable dosing based on serum levels

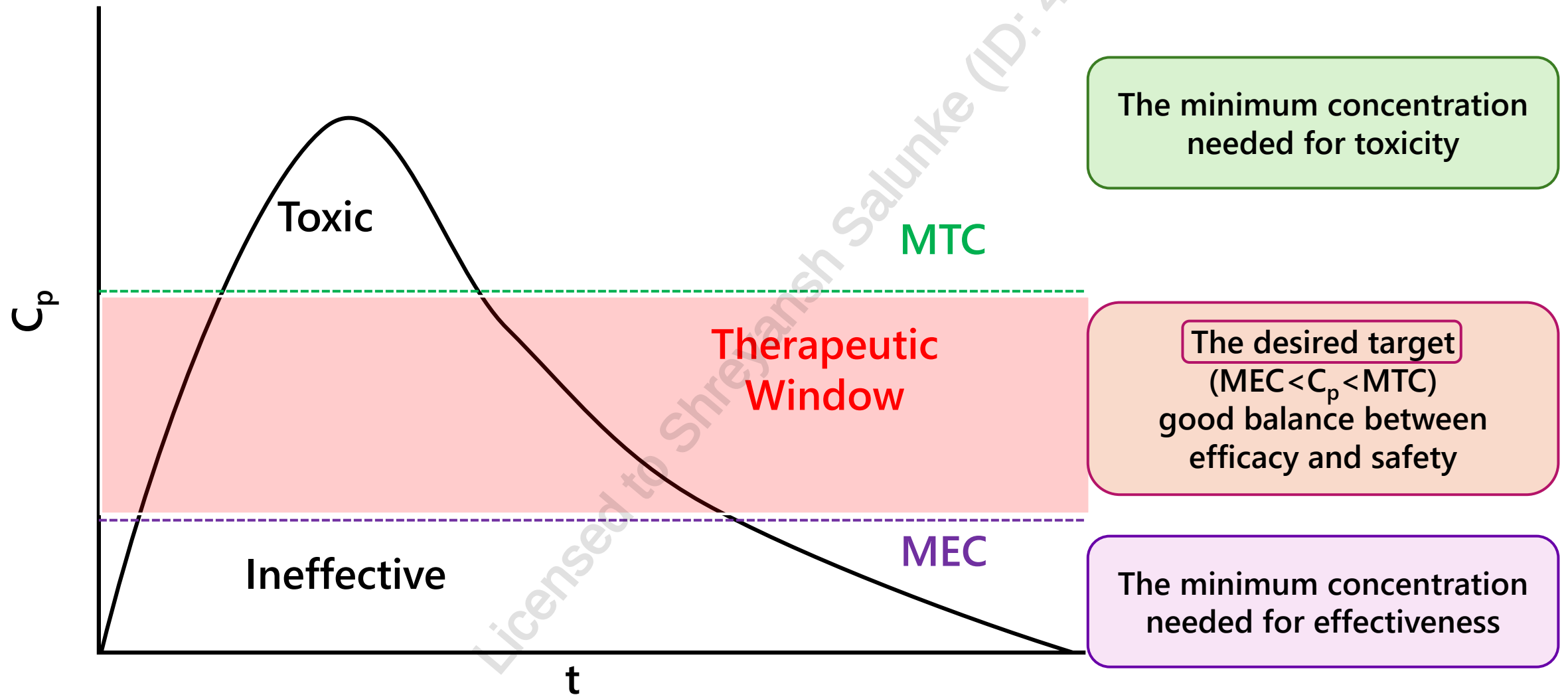
PD monitoring alone insufficient

Monitoring trough levels alone is acceptable after initial peak & trough calculation

Examples of drugs subject to TDM:

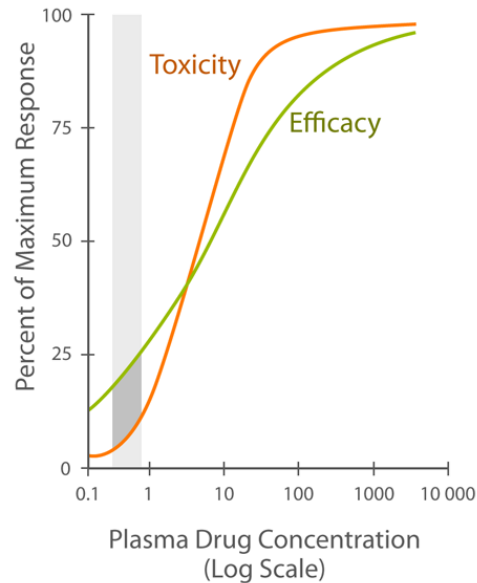
Gentamicin
Carbamazepine
Phenytoin
Valproic acid
Clozapine
Lithium

THERAPEUTIC WINDOW



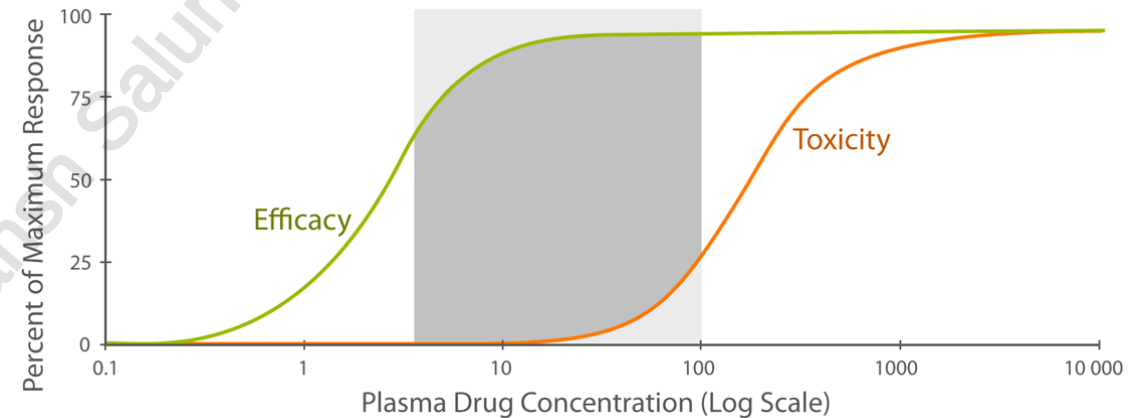
THERAPEUTIC WINDOW

Narrow Therapeutic Window



Maximal drug efficacy overlaps with toxicity to some extent; difficult to maintain values

Wide Therapeutic Window

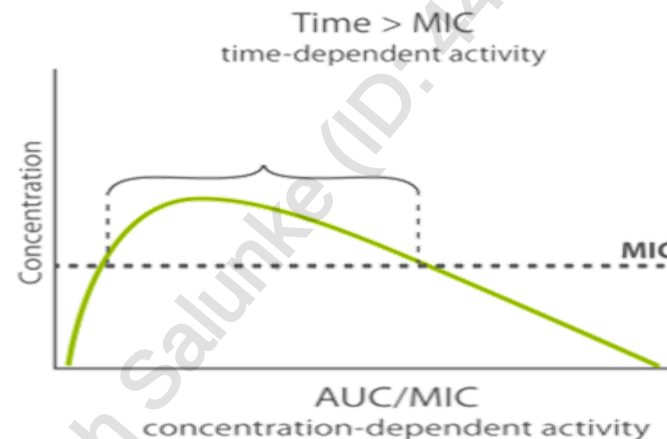


Maximal drug efficacy does not overlap with toxicity as toxicity occurs at much higher concentrations

ANTIBACTERIALS – PK/PD SUMMARY

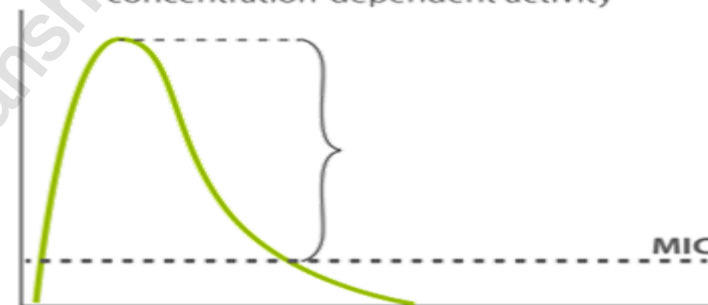
Time-dependent killing

- Bacterial killing (rate & extent) determined by **duration of time** that drug concentration at site of infection remain **above MIC**



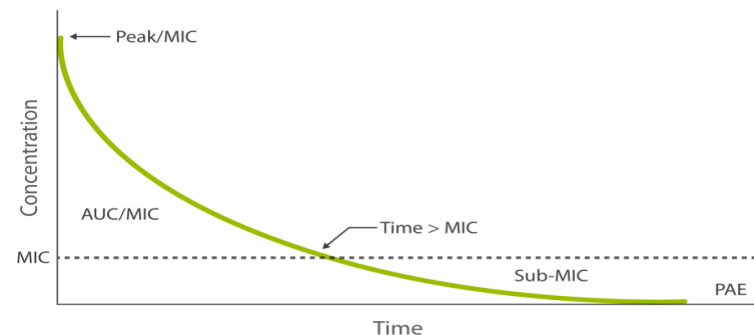
Concentration-dependent killing

- Bacterial killing (rate & extent) determined by **peak drug levels** or AUC/MIC ratio (i.e. exposure)



Post-antibiotic effect (PAE)

- Continued suppression of normal growth with antimicrobial levels **below** the MIC



Common Examples:

- Beta Lactams
- Macrolides
- Vancomycin
- Quinolones
- Aminoglycosides
- Aminoglycosides

AMINOGLYCOSIDES – CHARACTERISTICS

Mechanism of Action	Inhibit bacterial protein synthesis through irreversible binding to the 30S bacterial ribosomal subunit
Uses and monitoring	• For gram-negative infections: • Once daily dosing • Achieve high peak serum levels • For gram-positive infections (synergy): • Dosing q8-24 hours depending on pathogen and renal function • Monitor trough and peak levels
Dosing	Extend-interval or conventional
Pharmacokinetics	• Concentration dependent killing, post-antibiotic effect • Poor oral bioavailability (IV/IM only) • Volume of distribution: 0.2-0.3 L/kg • Almost exclusively renally cleared (~99%)
TDM	Extended-interval → Hartford nomogram and single serum concentration Conventional → peak and trough monitoring • Used with extended-interval not appropriate (pregnant patients, severe renal impairment, cirrhosis, burns)

AMINOGLYCOSIDE TDM: RENAL DYSFUNCTION

- NEPHROTOXIC!** Associated with renal tubular necrosis
- Patients with healthy renal function: $t_{1/2} \sim 2$ hr
 - Patients with chronic kidney disease: $t_{1/2} \sim 30-60$ hr

How to maintain steady-state peak and trough concentrations? **Extend interval between doses** (see table below)

Target peak level of 5-10 mg/L and trough level of 2.5 mg/L (gentamicin, netilmicin and tobramycin)
Target peak level of 15-20 mg/L and trough level of 5 mg/L (amikacin)

Drug	Dose (mg/kg)	CrCl >60 mL/min	CrCl 40-59 mL/min	CrCl 20-39 mL/min	CrCl <20 mL/min
Amikacin	15	Every 24 hr	36 hr	48 hr	7.5 mg/kg every 48 hr
Gentamicin	5-7				Load 2.5 mg/kg then 1.7 mg/kg every 48 hr
Netilmicin					
Tobramycin					

VANCOMYCIN - CHARACTERISTICS

Mechanism of Action	Inhibits bacterial cell wall synthesis via disruption of transglycosylation
Indications / Uses	- Gram-positive organisms (e.g., MRSA); IV <i>C. difficile</i> infection (only PO)
Dosing	Initial doses of 15-20 mg/kg, usually every 8-12 hours (however, interval may be longer depending on renal function)
Pharmacokinetics	- Time-dependent (or concentration-independent) killing, some PAE 2-compartment model - Poor oral bioavailability (100% IV bioavailability) - Volume of distribution: ~0.7 L/kg - Mostly renally cleared - Penetrates poorly into lung, CNS and bone due to large size
TDM	- Target trough (10-20 mg/L) or AUC:MIC of 400-600 - First order kinetics, dose adjustment generally proportional to trough level

CYCLOSPORINE TDM

Immunosuppressive calcineurin inhibitor with several indications (e.g., prevention of graft rejection, psoriasis, RA, nephrotic syndrome)

In most patients → **whole-blood 12-hour trough** (C_0) concentrations are monitored.

Target concentrations:

- 200 to 300 ng/mL in months 1 to 3 after transplantation
- 50 to 150 ng/mL for subsequent months

In patients suspected of having poor cyclosporine absorption, monitor **two-hour post dose** cyclosporine (C_{2hr} ; C2) levels should be measured.

Target concentrations:

- 800 to 1000 ng/mL in months 1 to 3 after transplantation
- 400 to 600 ng/mL for subsequent months

Higher C2 concentrations associated with decreased acute rejection rates in the first-year post-transplant

- However, C2 monitoring is often **more difficult and less convenient** for the patient

DIGOXIN TDM

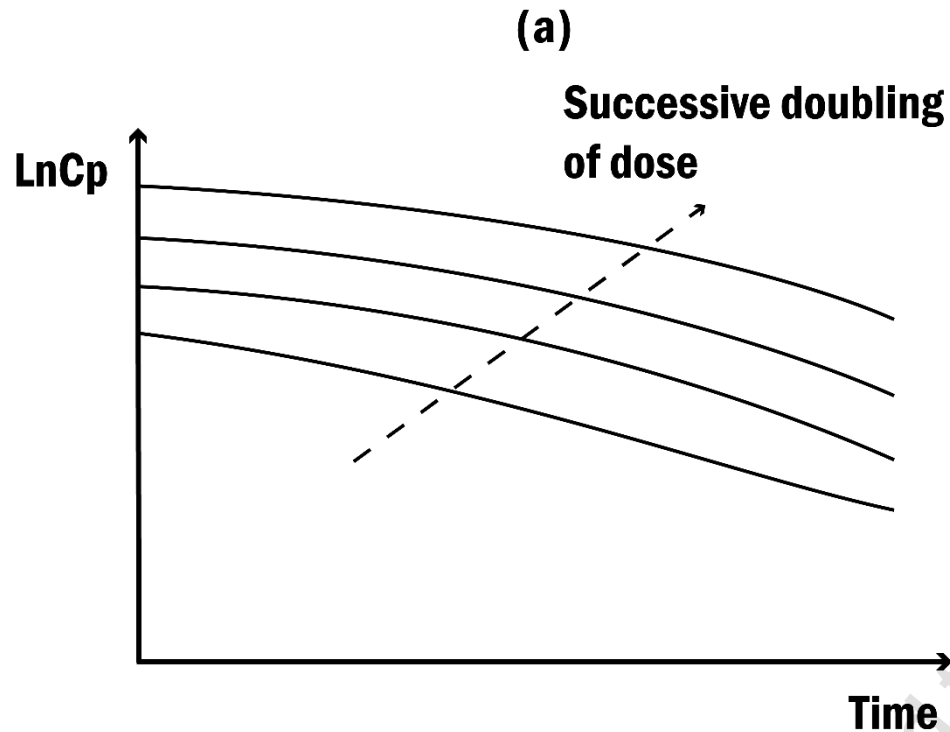
- Sodium/potassium ATPase inhibitor with **positive inotropic** and **negative chronotropic** effects; indicated for mild-moderate HF, control of ventricular rate in patients with chronic atrial fibrillation
- **Narrow therapeutic window** with potential for dysrhythmia and sinus bradycardia toxicities
- Several considerations for dosing digoxin: weight, **renal function**, age, comorbidities, **interactions (i.e., P-gp)**
- Monitoring:
 - Long half life (30-45h); levels should be measured 6-10 days after initiation of therapy
 - Sampling of serum concentrations should be done just before next scheduled dose of the drug
 - Levels should be drawn **no sooner than 6 hours after the most recent dose** (large Vd, distribution phase over 6-8 hours)
 - Therapeutic levels 0.6-1.2 nmol/L (recommended by Canadian Cardiovascular Society)

NON-LINEAR KINETIC DRUGS

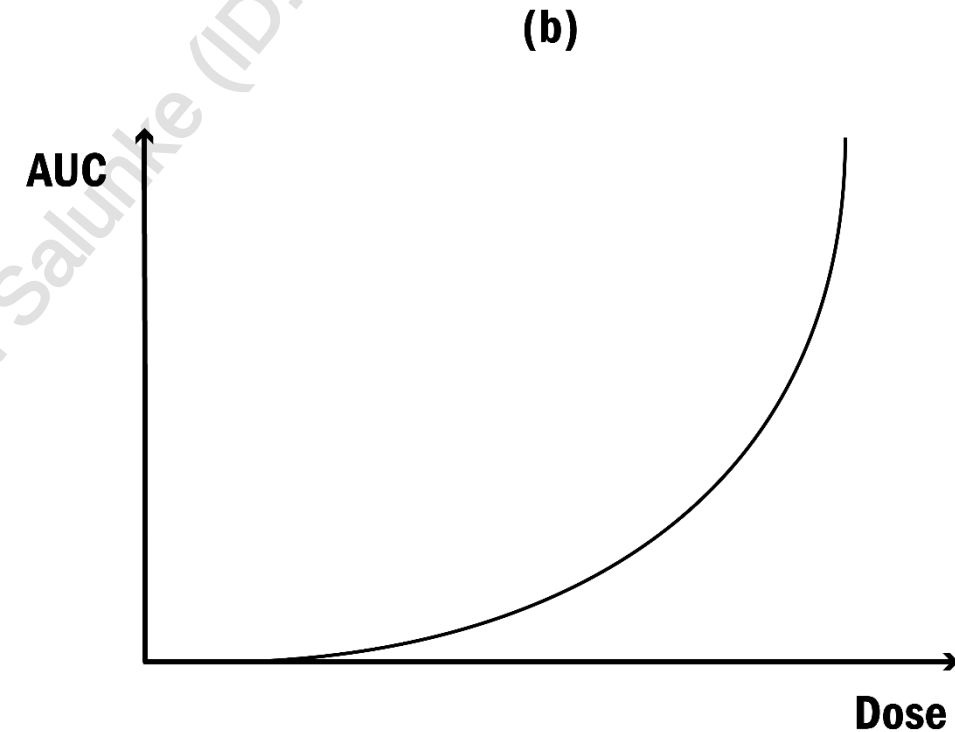
Linear kinetics	Non-linear kinetics
Rate of elimination proportional to C_p and Dose	Constant rate of elimination, irrespective of C_p or dose
Normal behavior for most drugs	Caused by saturation of transporters or enzymes
Changes in dosage parallels change in C_p and AUC; 2x increase in Dose = 2x increase in C_p and AUC	Change in dosage may have much bigger changes in C_p and AUC; 2x increase in dose > 2x increase in C_p and AUC
Variations in dosage is relatively OK if within the therapeutic window	Small changes in dosage can cause toxicity or loss of efficacy
Requires no special monitoring	Blood C_p should be monitored and renal correction may be used (Therapeutic Drug Monitoring)

NON-LINEAR KINETIC DRUGS

Impact of non-linear elimination on the plasma concentration–time profile after single doses.



Graph (a) is a semi-logarithmic plot of plasma concentration against time after a successive doubling of an intravenous dose. After large doses, the initial fall in the plasma concentration with time is less steep than with linear pharmacokinetics.



Graph (b) shows that the area under the plasma concentration–time curve (AUC) increases with dose for drugs that display saturable elimination.

SATURABLE KINETIC DRUGS

Drugs that are subject to saturable kinetics have a high risk of displaying non-linear kinetics

Drug	Uses and effect	Toxicity
Theophylline	Treatment of COPD Relaxes smooth muscle, ↑ respiration & HR	Nausea, vomiting, ionic imbalance Metabolic acidosis & tachycardia
Digoxin	Na ⁺ /K ⁺ ATPase inhibitor Increases cardiac contractility Treatment of arrhythmia	Tachycardia or fibrillation Nausea and vomiting Ocular disturbance
Procainamide	Rapid block of sodium channels Class IA antiarrhythmic	Immune system failure Pacemaker failure
Phenytoin	Sodium channel blocker Treatment of seizures and heart arrhythmias	Low BP Abnormal heart rhythms
Salicylic Acid	Active form of aspirin Commonly used NSAID	Primary metabolic acidosis Primary respiratory alkalosis Nausea, tachycardia, hyperthermia

PHENYTOIN

- Phenytoin is a commonly prescribed anticonvulsant and displays nonlinear PK
- Has a narrow therapeutic range and requires individualized dosing regimens
- Phenytoin is primarily eliminated by saturable metabolism → described by Michaelis-Menten equation
- Average K_m : 4 mg/mL
- Average V_{max} : 7mg/kg/day

ORAL ADMINISTRATION OF PHENYTOIN

The following equation can be used for phenytoin and other drugs that have similar saturable elimination:

$$F \cdot D = \frac{V_{max} \cdot Cp_{ss}}{Km + Cp_{ss}}$$

- V_{max} is the maximum metabolism speed achieved at enzyme saturation
- Km is the Michaelis-Menten constant
- Cp_{ss} is the steady-state plasma concentration
- F is the bioavailability
- D is the dose

PHENYTOIN STEADY STATE CONCENTRATION

The steady-state plasma concentration can be found by rearranging the previous formula:

$$Cp_{ss} = \frac{Km \cdot F \cdot R_a}{V_{max} - F \cdot R_a}$$

- V_{max} is the maximum metabolism speed achieved at enzyme saturation
- Km is the Michaelis-Menten constant
- Cp_{ss} is the steady-state plasma concentration
- F is the bioavailability
- R_a is the rate of drug administration; note $R_a = \frac{D}{\tau}$
 - D is the dose
 - τ is the dosing interval

PHENYTOIN MAXIMUM SPEED

The maximum speed of metabolism can be found by rearranging the previous formula:

$$V_{max} = \frac{(Km + Cp_{ss}) \cdot F \cdot R_a}{Cp_{ss}}$$

- V_{max} is the maximum metabolism speed achieved at enzyme saturation
- Km is the Michaelis-Menten constant
- Cp_{ss} is the steady-state plasma concentration
- F is the bioavailability
- R_a is the rate of drug administration

PRACTICE PROBLEM 10

Determine an intravenous loading dose of phenytoin sodium injection for AB, a male patient, who is 5'11", 71 kg, using a V_d of 0.65 L/kg and a desired level of 20 mg/L.

PRACTICE PROBLEM 10A

Determine an intravenous loading dose of phenytoin sodium injection for AB, a male patient, who is 5'11", 71 kg, using a Vd of 0.65 L/kg and a desired level of 20 mg/L.

Step 1: Calculate IBW

$$\text{IBW} = 50 \text{ kg} + 2.3 \text{ kg} * (\# \text{ of inches over } 5 \text{ ft})$$

$$\text{IBW} = 50 \text{ kg} + 2.3 \text{ kg} * (11)$$

$$\text{IBW} = 75.3 \text{ kg} \text{ (patient's actual weight} < \text{IBW} \rightarrow \text{use actual weight)}$$

Step 2: Calculate patient specific Vd

$$\text{Vd} = 0.65 \frac{\text{L}}{\text{kg}} (71 \text{ kg})$$

$$\text{Vd} = 46.15 \text{ L}$$

Step 3: Calculate dose

$$\text{Dose} = C * Vd$$

$$\text{Dose} = \frac{\left(\left(20 \frac{\text{mg}}{\text{L}} \right) * 46.15 \text{ L} \right)}{0.92}$$

$$\text{Dose} = 1003 \text{ mg}$$

PRACTICE PROBLEM 10B

Determine a daily maintenance dose of oral phenytoin sodium capsules for AB using a V_{max} of 10.18 mg/kg/day, K_m of 10.9 mg/L, $C_{ss} = 12.3$ mg/L. Round to the nearest dose using available phenytoin sodium capsules (100 mg extended-release capsules) to determine a daily dose for AB. Assume full bioavailability ($F=1$).

PRACTICE PROBLEM 10B

Determine a daily maintenance dose of oral phenytoin sodium capsules for AB using a V_{max} of 10.18 mg/kg/day, K_m of 10.9 mg/L, $C_{ss} = 12.3$ mg/L. Round to the nearest dose using available phenytoin sodium capsules (100 mg extended-release capsules) to determine a daily dose for the patient. Assume full bioavailability ($F=1$).

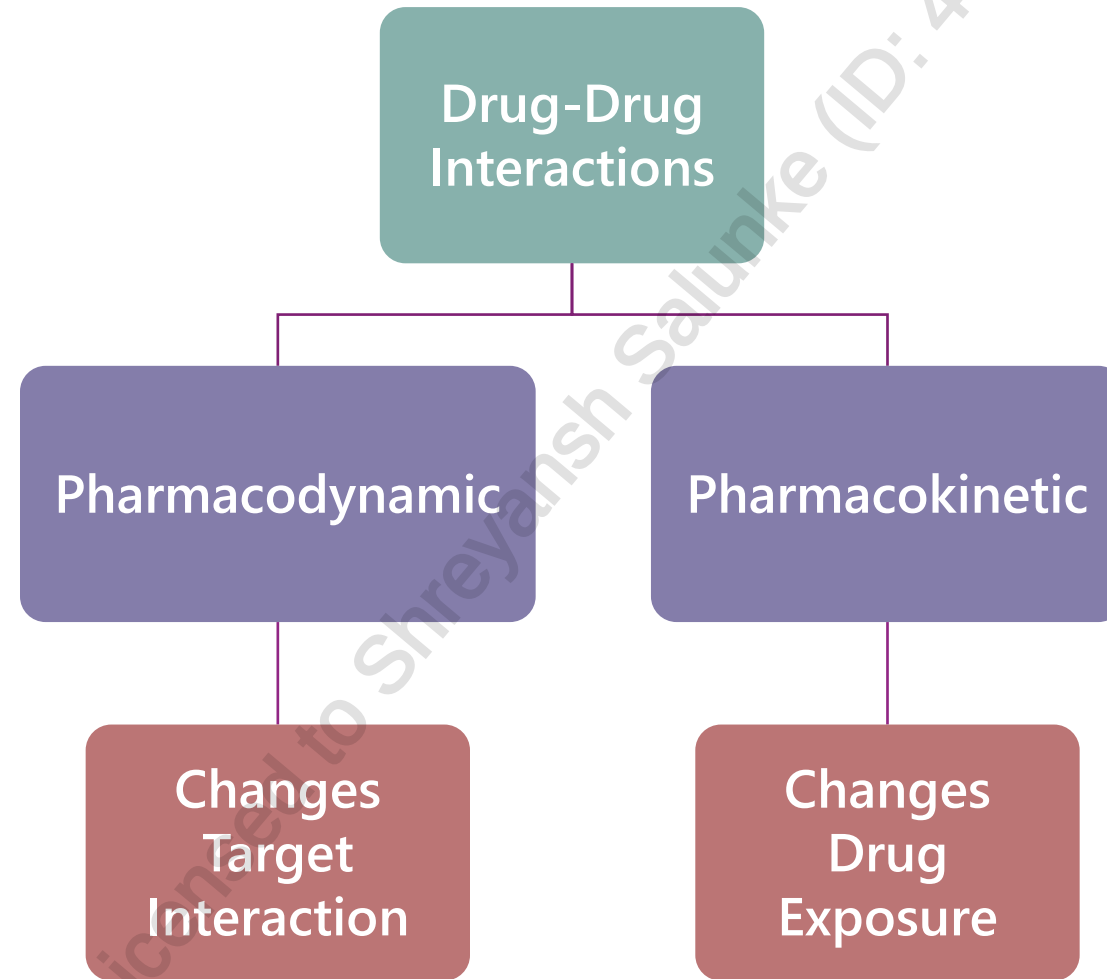
Step 1: Calculate patient specific V_{max}

$$71 \text{ kg} * 10.18 \text{ mg/kg/day} = 722.78 \text{ mg/day}$$

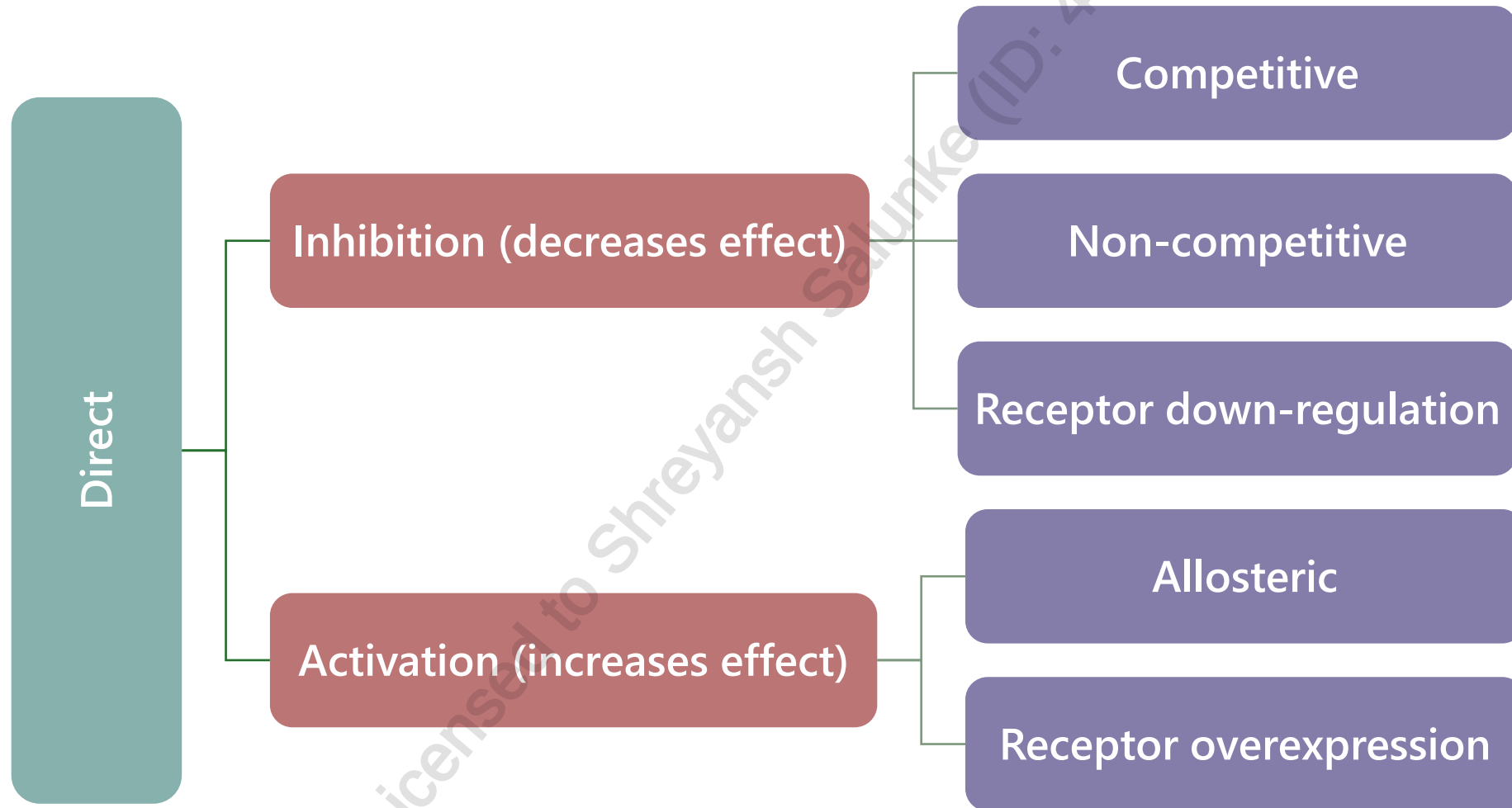
Step 2: Plug in givens

$$F \cdot D = \frac{V_{max} \cdot C_{p_{ss}}}{K_m + C_{p_{ss}}}$$
$$D = \frac{722.78 \cdot 12.3 \frac{\text{mg}}{\text{L}}}{10.9 \frac{\text{mg}}{\text{L}} + 12.3 \frac{\text{mg}}{\text{L}}} = 383 \frac{\text{mg}}{\text{day}} \sim 400 \frac{\text{mg}}{\text{day}}$$

KINETICS OF INTERACTIONS



KINETICS OF INTERACTIONS



PHARMACODYNAMIC INTERACTIONS

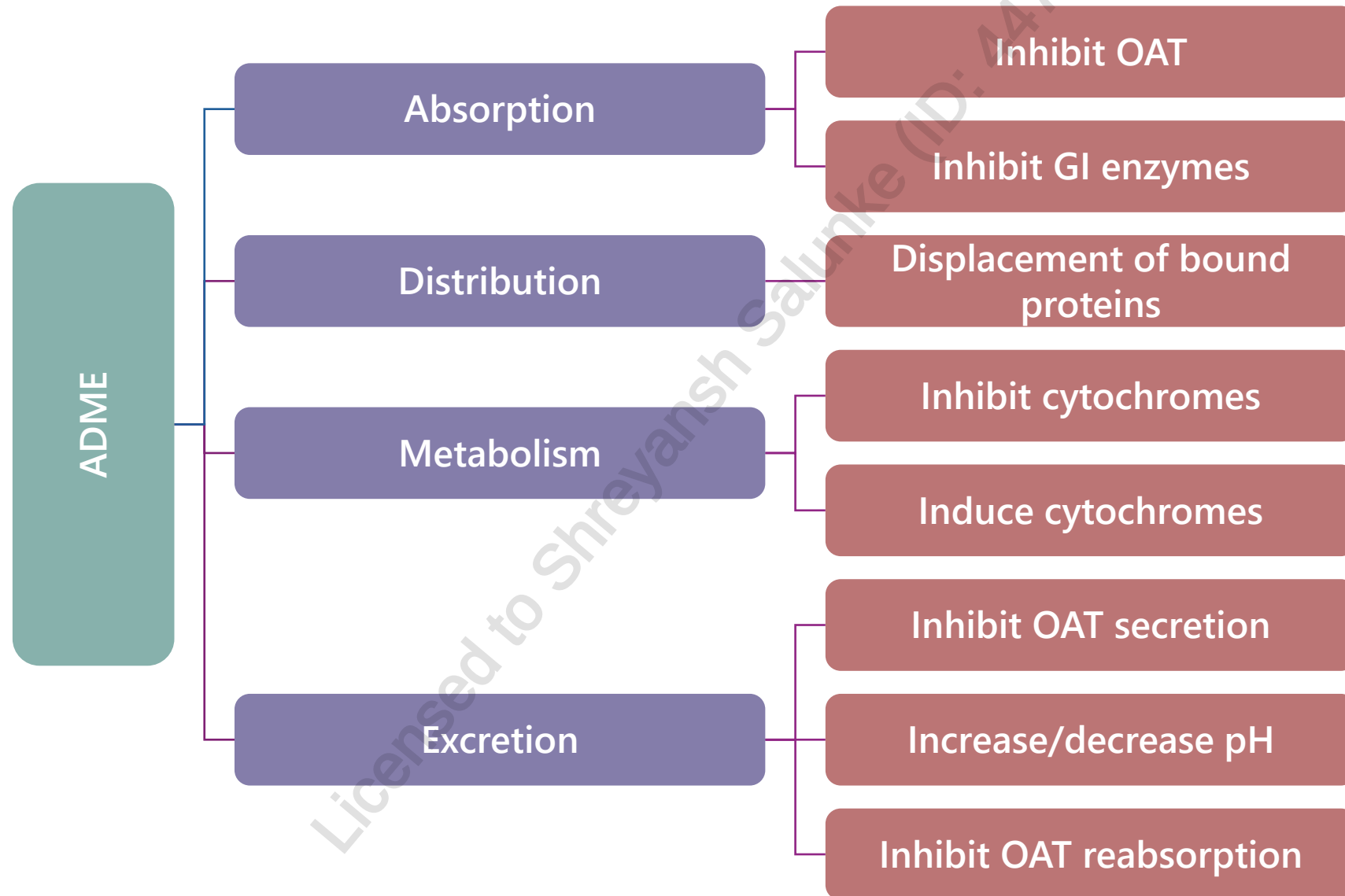
SSRI + MAOI

- Synergistic increase in serotonin levels causing serotonin syndrome; hypertensive crisis

Sildenafil + Nitroglycerin

- Synergistic increase in cGMP (sildenafil inhibits catabolism via phosphodiesterase type V, nitroglycerin donates nitric oxide to activate guanylate cyclase) to potentiate hypotensive effects

KINETICS OF INTERACTIONS



PHARMACOKINETIC INTERACTIONS

6-Mercaptopurine/Azathiopurine + Allopurinol

- Allopurinol inhibits xanthine oxidase-mediated metabolism of purine analogues (e.g., 6-mercaptopurine, azathioprine)

Caffeine + Ciprofloxacin

- Ciprofloxacin inhibits CYP1A2-mediated caffeine metabolism causing restlessness, anxiety, insomnia

Digoxin + Amiodarone

- Amiodarone inhibits Pgp-mediated clearance of digoxin causing dysrhythmia, sinus bradycardia

Warfarin + Amiodarone

- Amiodarone inhibits CYP2C9-mediated metabolism of warfarin; potentiates the effect of warfarin and prolongs the International Normalized Ratio (INR); increasing the risk of bleeding

PHARMACOKINETIC INTERACTIONS

Carbamazepine + Macrolide antibiotics

- Macrolide antibiotics (e.g., clarithromycin, erythromycin) inhibits CYP3A4-mediated carbamazepine metabolism causing drowsiness, ataxia

Opioids + Azole antifungals

- Azole antifungals (e.g., ketoconazole, itraconazole) inhibit CYP3A4-mediated opioid metabolism causing respiratory depression

Opioids + Rifampin

- Rifampin induces CYP3A4-mediated opioid metabolism reducing analgesic effects

PHARMACODYNAMICS- HOW DRUGS AFFECT THE BODY

Factors affecting drug response

- Receptor site availability and expression
- Signal transmission via second messengers
- Gene regulation, translation and protein production
- Tolerance
- Drug interactions at the receptor level: Synergistic, additive, antagonistic

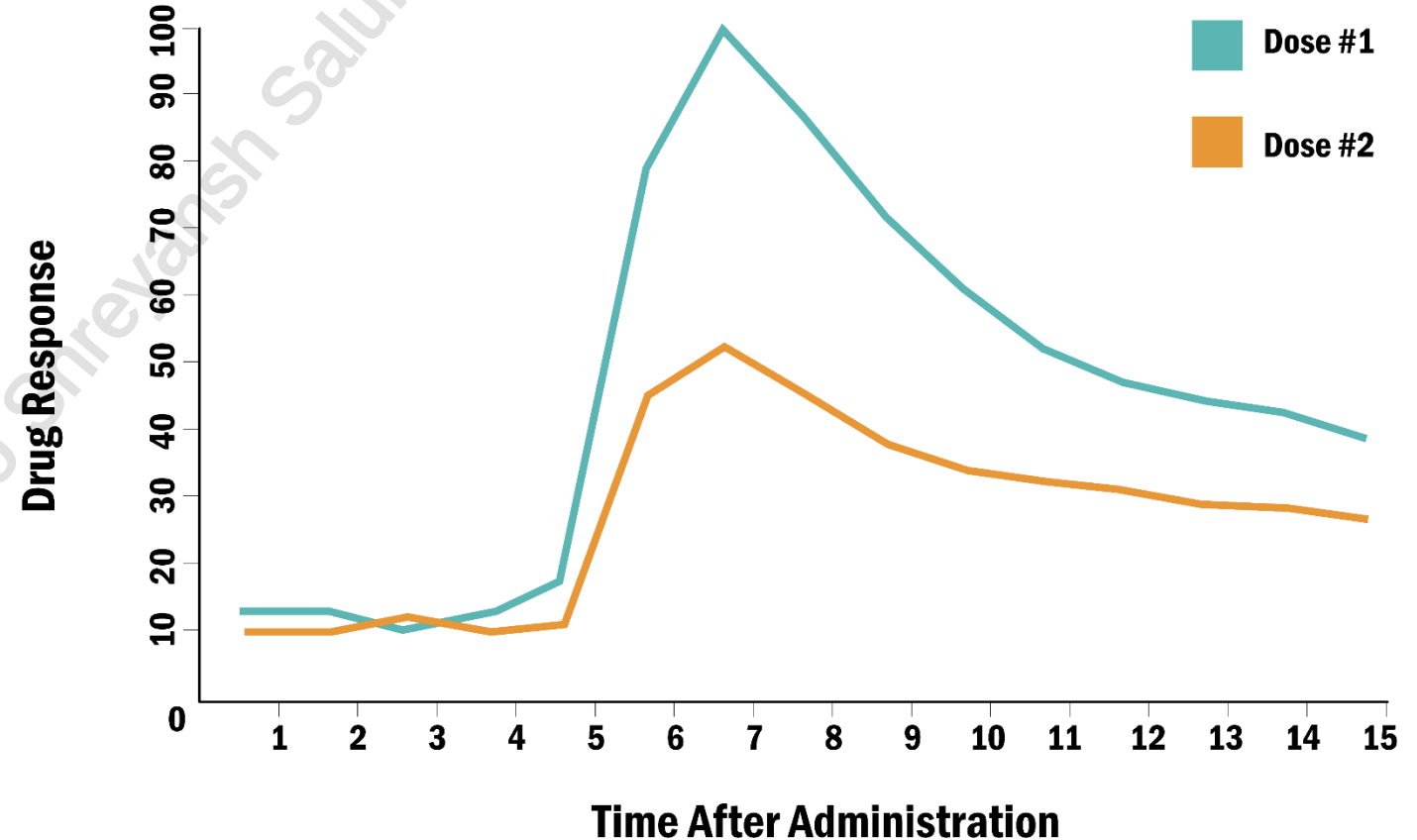
EC_{50} : The effective concentration (EC) of the drug where 50% of the maximum drug effect is seen (Dose-response log scale curves)

- It is a parameter used to compare drug potency of several drugs
- The lower the EC_{50} → the more potent the drug
- Does not indicate duration of effect which is determined by intracellular signaling and gene regulation systems (2nd messengers)

PHARMACODYNAMICS

- Tolerance can be seen using drug effect plasma concentration curves (log scale)
- Same dosing gives you **less effect or higher EC50**, seen with continued use of drug (for same effect → need higher doses)
 - Due to **pharmacokinetic** factors such as drug metabolism
 - Example: Auto-induction causes decreased drug concentration
 - Due to **pharmacodynamic** factors – same concentration at the receptor site causes less drug effect
 - Example: pain management with opioids

Tolerance: The decreased efficacy of a drug with repeated use as the body adapts to the continued presence of the compound.



SUMMARY

Part 1 (ADME)	Part 2
• Absorption: how drug enters systemic circulation from the site of administration, many factors can affect rate/extent (e.g. solubility, pH, formulation, etc.) • Distribution: how drug distributes through body fluids and tissues (i.e. Vd), affected by blood, protein binding • Metabolism: making drug easier to eliminate, Phase 1 (oxidation, reduction, hydrolysis) and 2 (conjugation) reactions • Elimination: removal of drug from urine or feces	• 1-compartment model often used in PK to help simplify the analysis and interpretation of drug behaviour • Drug therapy often requires <u>multiple/repeated</u> dosing for patient maintenance. Some considerations: • Fluctuation • Accumulation • Steady state • Steady state achieved in ~5 half-lives of a drug (though a loading dose can be used to achieve maintenance levels quickly) • TDM needed for drugs where their behaviour can be influenced by multiple patient-specific factors, narrow therapeutic window • Linear PK – drug concentration proportionally changes with dose vs. non-linear PK is disproportionate due to saturable processes (e.g., phenytoin) • Pharmacodynamics – how drug affects the body (i.e. MoA, therapeutic and side effects)

PHARMACOKINETIC EQUATIONS

Equation	Comments
$C_t = C_0 \cdot e^{-k_{el} \cdot t}$	C_t = concentration at time t (mg/L) C_0 = concentration at the end of the infusion (mg/L) t = any time after the end of the infusion (hours)
$C = C_0 * 10^{-\frac{kt}{2.303}}$	C = concentration ; C_0 = initial concentration k = rate constant T = Time at concentration "C"
$C_{min} = C_t \cdot e^{-k_{el} \cdot t_2}$	C_{min} = extrapolated trough concentration (mg/L) C_t = measured concentration (mg/L) t_2 = time from measured trough to start of next dose (h)
$\log C = \log C_0 - \frac{k}{2.303} t$	C = concentration ; C_0 = initial concentration k = rate constant T = Time at concentration "C"
$1 - e^{-k_{el} \cdot t}$	Fraction lost during decay or fraction achieved during the infusion
$C_{max} = C_p \cdot e^{-k_{el} \cdot t_1}$	C_{max} = extrapolated peak concentration (mg/L) C_p = measured peak concentration (mg/L) t_1 = time from end of infusion to measured peak (h)

PHARMACOKINETIC EQUATIONS

Equation	Comments
$V_{ss} = \left(\frac{\text{Dose}}{k_{el} t_i} \right) \left(\frac{(1 - e^{-k t_i})}{C_{max} - (C_{min} \times e^{-k t_i})} \right)$	V_{ss} = volume of distribution at steady-state (L) Dose = administered dose (mg) t_i = infusion time (hours) k_e = elimination rate constant (h^{-1}) C_{max} = extrapolated peak concentration (mg/L) C_{min} = extrapolated trough concentration (mg/L)
$t_{1/2} = \frac{0.693 \times V_{ss}}{Cl_{ss}}$	$t_{1/2}$ = half-life (hours) Cl_{ss} = clearance of drug in steady-state (L/h) V_{ss} = volume of distribution at steady-state (L)
$\text{New } C_{tss} = \text{New } C_{pss} \cdot e^{-k_{el} \cdot T}$	New C_{tss} = trough concentration at SS (mg/L) k_{el} = elimination rate constant (h^{-1}) $T = n\tau - (t_i + t_1)$ (h)

PHARMACOKINETIC EQUATIONS

Equation	Comments
$Cl_{ss} = k_{el} \cdot V_{ss}$	Cl_{ss} = clearance of drug in steady-state (L/h) k_{el} = elimination rate constant (h^{-1}) V_{ss} = volume of distribution at steady-state (L)
$k_{el} = Cl_{ss}/V_{ss}$	Cl_{ss} = clearance of drug in steady-state (L/h) K_{el} = elimination rate constant (h^{-1}) V_{ss} = volume of distribution at steady-state (L)
$E_{\tau} = \frac{\ln(C_{pd}/C_{td})}{k_{el}} + (t_i + t_1)$	E_{τ} = estimated new dosing interval (hours) k_e = elimination rate constant (h^{-1}) C_{pd} = desired peak concentration (mg/L) C_{td} = desired trough concentration (mg/L) t_i = infusion time (hours) t_1 = time from the end of infusion to measure peak (hours)

PHARMACOKINETIC EQUATIONS

Equation	Comments
$\text{New Dose} = t_i \times CL_{ss} \times \frac{C_{pd}}{e^{-kt_1}} \times \frac{1 - e^{-kn\tau}}{1 - e^{-kt_i}}$	New Dose = estimated new dose (mg) T_i = infusion time (hours) K_e = elimination rate constant (h^{-1}) CL_{ss} = clearance of drug at steady-state (L/h) C_{pd} = desired peak concentration (mg/L) t_1 = time from the end of infusion to measured peak (hours) $n\tau$ = new dosing intervals (hours)
$A = \text{Dose} \cdot e^{-k_{el} \cdot t}$	A = amount of drug remaining in the body Dose = dose k_{el} = elimination rate constant t = time $t_{1/2}$ = elimination half-life

PHARMACOKINETIC EQUATIONS

Equation	Comments
$C_{ss} = \frac{R_0}{CL}$	C_{ss} = blood concentration at steady-state R_0 = the infusion rate CL = whole-body clearance
$A_{ss} = \frac{R_0}{k_{el}}$	A_{ss} = amount of drug in the body at SS k_{el} = the elimination rate constant
$\begin{aligned} &\text{New } C_{p_{ss}} \\ &= \left(\frac{\text{New dose}}{t_i} + CL_{ss} \right) \left(\frac{1 - e^{-k_{el} t_i}}{1 - e^{-k_{el} N\tau}} \right) \cdot e^{-k_{el} \cdot t_1} \end{aligned}$	New $C_{p_{ss}}$ = predicted peak concentration at steady-state (mg/L) New Dose = chosen new dose (mg) t_i = infusion time (hours) k_{el} = elimination rate constant (h^{-1}) CL_{ss} = clearance of drug at steady-state (L/h) t_1 = time from the end of infusion to measured peak (hours) $N\tau$ = new dosing intervals (hours)

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CHANGE LOG

August 2025

- Text spelling and formatting changes throughout
- Added s66-83 from part 1 as continuation
- Slide 88 - added 'rate in= rate out', removed "proportional to bioavailability" since redundant (slide on IV infusion)
- Slide 93 - fixed missing superscript in answer
- Slide 97 – reworded/simplified text under "considerations"
- Slide 99 - removed subscript 'n' in equation, as it is steady state equation
- Moved 'multiple doses figures' slide to slide 104 after 'fluctuation'
- Slide 108 - altered title to 'loading vs maintenance doses', added "and rapidly achieve therapeutic plasma concentration" under Loading Dose
- Slide 109 - removed subscript beside 'k'
- Slide 119, 121– organized information into table and aligned with information in TDM deck, slide 124 hidden (can speak verbally with s123)
- Slide 126 - reformatted for visual clarity
- Slide 133 - fixed answer/calculations to use actual body weight (since actual < IBW)
- Slide 140 - removed digoxin/quinidine interaction as quinidine not available in Canada, fixed digoxin/amiodarone as interaction is due to Pgp

October/November 2025

- Slide 74: adjusted wording and added text box for clarity re: why some IV drugs can be modelled using 1-compartment kinetics
- Slide 75, 99, 100: removed 'F' as redundant, equations relate to IV
- Slide 109: added 5 half-lives = 97.25%

- Slide 133: Added male gender to case
- Slide 144: added summary slide

Feb 2026

- Added slides 6-12 (compartment models) as was not covered in Part 1
- Slide 66: minor edit